VDI|VDE|IT

Study

on the

National Innovation System in Jordan

Final Report

VDI/VDE
Innovation + Technik GmbH

Uwe Seidel
Dr. Wolfgang Domröse
Dr. Gerd Meier zu Köcker

Steinplatz 1
10623 Berlin

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On behalf of

gtz

Federal Ministry
for Economic Cooperation
and Development

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1 Preface

1.1 Concept of National Innovation Systems

The competitiveness of nations and regions is nowadays not determined by single companies, but more and more by the innovative activities of entire industries and branches. For this reason, regional and national competitiveness has become the central topic for the economic and technology policy world-wide. Innovative firms grow faster and are more likely to survive during a recession. The benefits of innovation are not only limited to the original innovator though; by its own nature the production of innovation creates knowledge spill-overs that allow other firms to benefit from the initial innovation in terms of increasing total factor productivity. In turn this can create the condition for a virtuous circle of economic growth from which the whole of society can benefit.
Stimulating research and innovation can be considered as one of the major tasks of a national approach to increase the wealth of a domestic economy. It aims at different objectives, like economic growth, full employment, stable prices or positive balance of payments from international trade. Insufficient innovation was recently considered a major cause of Europe's disappointing growth performance[1]. Not surprisingly, then, stimulating the innovation performance of European member states is one of the main objectives of respective policy-makers. Nowadays, innovation is also of high priority in the Middle Eastern countries and surrounding areas. Plenty of policy measures and support schemes aimed at innovation have been implemented or are under preparation, with different impact. The diversity of these measures and schemes reflects the diversity of the framework conditions, cultural preferences and political priorities in the Middle East. Policy makers can stimulate the innovation ability of a national innovation system (NIS) by setting appropriate framework conditions as well as by public investments in infrastructure, education and funding innovation stimulating programmes.

All these measures and related effort aim to improve the performance of a national innovation system (NIS). Although there is no harmonised definition, these both options try to better explain what is meant by a NIS

- *.. the network of institutions in the public and private sectors whose activities and interactions initiate, import, modify and diffuse new technologies*[2]

- *.. the elements and relationships which interact in the production, diffusion and use of new and economically useful knowledge ... and are either located within or rooted inside the borders of a nation state*[3]*.*

[1] Expert group of the European Commission (chaired by former Prime Minister of Finland Esko Aho), Report “Creating an Innovative Europe”, 2006

[2] Freeman, C., “The National System of Innovation in Historical Perspective”, Cambridge Journal of Economics, No. 19, pp. 5–24, 1995

[3] Lundvall, B-Å. (ed.), National Innovation Systems: Towards a Theory of Innovation and Interactive Learning, Pinter, London, 1992

There is no formal common organisational framework for a NIS, neither in the Middle East nor elsewhere in the world. The main elements of a NIS in terms of education and research institutes, firms, industrial parks, incubators, governmental institution, etc. exists, but differs in terms of how they are coordinated or meshed. The key players in a national innovation concept typically include:

- Governmental and public authorities
- Firms
- Educational entities
- Research institutions
- Intermediaries
- Banks and financial institutions
- Other enablers.

Innovations typically are the result of a complex set of relationships among actors in the Innovation System, which includes all actors mentioned above. Innovations are related to new technologies, products, processes and increasingly, new service. For policy-makers, a better understanding of their own NIS can help to identify leverage points for enhancing innovative performance and overall competitiveness of a nation. It can assist in pinpointing mismatches within the system, both among institutions and in relation to government policies, which can thwart technology development and innovation. Policies, which seek to improve networking among the actors and institutions in the system and which aim at enhancing the innovative capacity of firms, particularly their ability to identify and absorb technologies, are most valuable in this context. As a consequence clusters became more and more the focus of today's innovation supporting actions[4].

The concept of NIS rests on the premise that understanding the linkages among the actors involved in innovation is a key factor to improving technology performance. Innovation and technical progress are the result of a complex set of relationships among actors producing, distributing and applying various kinds of knowledge. The innovative performance of a country depends to a large extent on how these actors relate to each other as elements of a collective system of knowledge creation and in what extend they utilise technologies. These actors are primarily private enterprises, universities and public research institutes. The linkages can take the form of joint research, personnel exchanges, cross patenting, purchase of equipment and a variety of other channels[5].

[4] Meier zu Köcker, Gerd, Clusters in Germany , www.iit-berlin.de/veroeffentlichungen, 2008

[5] National Innovation Systems, OCED Report, 2005

1.2 Scope and Objectives of the Study

Although significant progress has been made in many cases, NIS in Northern Africa and the Near East are still seeking for efficiency and effectiveness to reach sustainable impacts on innovation and competitiveness. For policy makers the main challenge remains to set up appropriate measures to stimulate innovations, which are demand oriented and tailor-made with the financial investments available. Although it is well understood that stimulating innovations and improving the NIS is a task of public authorities, these still hesitate to really commit themselves and significantly invest in related measure and infrastructure. One of the main reasons is that a NIS appears to be very complex and influenced by many determinants. Thus, it is very difficult for policy makers to decide where or how to start.

The main objective of this study is to analyse the NIS of Jordan according to its main determinants. It is quite clear that other studies on the NIS of Jordan already exist, including recommendation for improvements. But these studies and the corresponding recommendations appear sometime very generic or too general for policy makers to turn them into practice. Voluminous reports and plenty of recommendations are the outcome, which overloads policy makers in these countries, who are not that experienced in innovation policy matters. Knowing that their NIS needs to be improved, the crucial questions for policy makers are: where and how to start, even if the capability for public investments is low, where can the Deutsche Gesellschaft für Technische Zusammenarbeit (GTZ) GmbH provide some limited input generating maximum impact? These questions will be answered by means of this study and the methodology used.

The GTZ defines a NIS to be the interaction between the four pillars of technological capabilities. These pillars are:

- The political, legal regulatory framework that stimulates learning and innovativeness;
- The education system which includes the comprehensive primary education, technology-related secondary education, vocational training, higher education and continuing education at all levels;
- Technology-related and R&D institutions, whose roles include intellectual rights protection, basic research, R&D financing, technology transfer and extension, technology assessment, technology information and dissemination, consultancy firms and testing, measuring, quality control and standards certification;
- Innovation capability at firm level, which includes the existence of creativity of employees at all levels, innovation-friendly organisational structure and networked research and development.

In short the GTZ understands that the NIS is a network between the state with its provided frame conditions, the technology-related and R&D institutions, the education system and the industry level.

The study focused on the following subjects and tasks

- To analyse the existing strategies for industrial development as well as SME and innovation policy
  - Get a better understanding of the NIS
  - Identify missing linkages/ challenges in the NIS
  - Identify missing linkages between R&D institutions and other entities
- To develop different options for the promotion of innovation and technology in consideration of existing support programmes
  - Give an overview available policy tools
  - Develop different recommendations as foundation for decision-making in terms of possible interventions and funding priorities

The study is structured as follows: The research results are documented and discussed in chapter 4; recommendations for future intervention are made in chapter 5. In addition, a comparative overview concerning the main differences between the NIS of Syria, Jordan and Egypt are defined and discussed in chapter 6. Selected examples of innovation related literature are described in the Appendix.

1.3 Economic Situation in Jordan

Located at the intersection of Europe, Asia and Africa, the Hashemite Kingdom of Jordan is bordered by Israel to the west, Saudi Arabia to the south, Iraq to the east and Syria to the north.

Jordan is notably resource-poor, with limited agricultural land, no oil resources and considerably scarce water. Its only natural resources are potash and phosphate. Employment is mainly based on services (77%). 20% of the labour force depends on industry and 3% on agriculture. The Jordan economy grew by about 6% per year during the period 2006 to 2008[6]. It is estimated to remain high at about 5.5% during the next years, mainly led by manufacturing, construction, real estate and services sectors[7]. The main sources of the GDP are shown in figure 1.

[6] CIA Factbook, 2009

[7] http://go.worldbank.org/GIGQMWXLX0

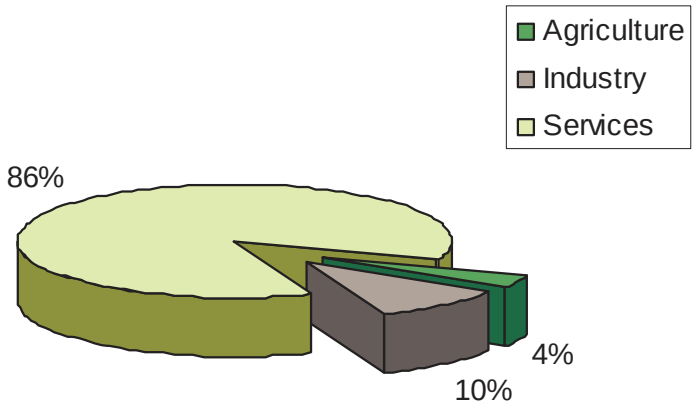


Figure 1 Origin of GDP according to sector[6]

Based on The Global Competitiveness Report 2008-2009, some of the recent findings on Jordan's competitiveness are summarised[7]. Well-defined property rights, efficient government spending, a low burden of government regulation and a fairly efficient legal framework, coupled with a very safe and secure environment ensure that the country's institutions receive a positive assessment. At the same time, the country's macroeconomic situation is quite weak. According to the global ranking, Jordan fell down by 11 positions. Major concerns are the growing budget deficit. Moving forward, Jordan is called up to also address its low primary education enrolment rate as well as to make the labour market more flexible. These actions would have a positive impact for the country's business sector and employment creation.

Further improvements in the investment environment will help advance private sector development in Jordan. It is acknowledged that perceptions of corruption, security and property rights became much better during recent years. In addition, the financial sector has been able for the most part to meet the needs of operating enterprises and infrastructure is not a major constraint. Jordan is nowadays focusing on addressing some of the remaining constraints, including streamlining civil proceedings, reforming taxation and tax administration, providing demand-driven skills training for workers, providing greater opportunities for private employment agencies to operate in Jordan's labour market and developing the capacity of insolvency administrators.

Using the demographic opportunity of a very young population to transform Jordan from a small lower-middle income and vulnerable country into a modern knowledge-based economy is at the core of the countries vision. This implies a bold modernisation of the country's economic, institutional and political infrastructure, based on the enhancement of its human capital and the elimination of poverty. Education, improving the business environment and attacking deep poverty are at the core of this vision.

Jordan is opening the economy much more than neighbouring countries through the establishment of a network of free trade agreements (FTAs) with strategic commercial partners. Jordan recently signed FTAs with the USA (its main export market) and Turkey. It is a member, along with Egypt, Morocco and Tunisia, of the Agadir FTA and it is moving towards the dismantling of tariff barriers on imports from the EU by 2011.

Jordan signed the *Euro-Mediterranean Charter for Enterprise* in October 2004. The 2008 report on the implementation of the charter shows, that over the last 10 years, Jordan has been very active in reforming its economy. It ranks as one of the best reformers compared to other middle income-countries. Despite the progress made, Jordan still confronts several challenges, including those that will arise from the recent global economic slowdown. Challenges include vulnerability to fluctuations in the international oil market, due to the country's high energy import dependency, high unemployment and dependency on remittances from Gulf economies and increasing pressure on natural resources, especially water. The greatest challenge (and also the largest opportunity) remains the necessity to create adequate conditions for increased private investment and improved competitiveness. This will help to deliver the high and sustainable growth needed to create employment and to reduce poverty. Staying the course with the implementation of the fiscal consolidation programme may prove more challenging in the future but this is one major key to preserve good economic performance.

1.4 Acknowledgements

The study was carried out between August and September 2009 by VDI/VDE-IT in close cooperation with GTZ and local partners.

One stay in Jordan by VDI/VDE-IT experts served the purpose of conducting a series of interviews with actors within both innovation systems, consisting mainly of representatives of ministries, representatives of universities, technology centres, innovation support programmes and stakeholder organisations from industry. The complete list of interviewees is given in Appendix II.

We are very grateful to the GTZ-colleagues in Jordan who strongly supported our work. We would also thank our network partners in these countries for their assistance.

Berlin, October 2009

2 Methodology

Theoretic models and descriptions of NIS as well as their analyses have been increasing in number since the early 2000s. Assessing a country's innovation system is a challenging task because of the enormous number and variety of influences on national innovation capability. For years, economists have tried to understand what determines the performance of the competitiveness of nations, the outcomes of innovation approaches and the wealth of nations. As a consequence, many reports and analyses of NIS have been generated. Although these reports were excellent, described the main elements of a NIS and caught its main strengths and weaknesses, the benefits in terms of usable results were mostly limited. The main reason is that the applied methodologies of the studies were not perfectly linked with the circumstance how policy makers think and operate. Recommendations are neither prioritised nor ranked according to their complexity when turning into practice.

Policy makers, especially in emerging and developing countries, usually are looking for well structured descriptions of a NIS and clear recommendations how to improve the functionality of a NIS and what kind of specific measures were recommended. They do not ask for receiving scientific models of the functionality of a NIS or how the single actors are linked. As far as embryonic or not well established NIS are analysed, they have mainly been compared with those that are matured. The consequences are plenty of weaknesses found and recommendations made. Often policy makers are confused rather than getting a clear guideline on how to start corrective actions. Such reports have often failed to provide clear information or recommendations how to start and how to gain a high leverage effect (especially when public investments were limited). When it comes to GTZ concerns, it is of main interest to identify those determinants of a national innovation system, which can be improved by the tools and financial means available for GTZ.

The present report falls into the new tradition of indicator-based studies based on quantitative data generated by the evaluation of expert interviews. The Global Competitiveness Report or the European Scoreboard on Innovation is an excellent approach to benchmarking the performance of NIS. The latter one is mainly intended for well matured economies and NIS and cannot sufficiently be applied for developing or emerging countries since the statistical base is often insufficient. The Global Competitiveness Report uses a mix of statistical data and expert interviews, but it is focussing on the competitiveness of nations.

The overall approach we have applied in this study can be divided in the following steps:

- Analysis of literature on NIS
- Conducting interviews with experts on NIS
- Evaluation and measuring the results
- Identifying those determinants which appear to high impact with less efforts
- Deducting recommendation how to improve the prioritised determinants

The study captures this challenge by providing an indicator based assessment of many different determinants, each of which reflects one aspect of the complex reality that we call innovation. We

grouped all these determinants according to the macro, meso and micro level figure 2 describes the different dimensions and its actors.

Level		Actors	Functionality within a NIS
Macro	Policy	Public authorities, policy makers	Governing and setting framework conditions of a 'NIS
Meso	Institutional innovation support Programmatic innovation support	Institutional innovation supporter or public funded initiatives / programmes	Institutions and initiatives are tools to turn innovation policies into practice
Micro	Innovation capacity	Firms, academia, educational institutions, etc.	Main beneficiaries of support measures and main producers of knowledge, innovation, technologies, products

Figure 2 Levels and actors within a NIS

Macro Level – Innovation Policy Level

On a macro dimension national and regional innovation policies directly influence the framework conditions of a NIS. Laws, decrees and regulations, etc. on that level are often path breaking in a positive or negative sense. The readiness of public investments in innovation is a direct outcome on decisions made on policy level. But, decision on this level can only influence the framework condition of a NIS and do not turn innovation into practice.

Meso Level - Institutional Innovation Support Level

Institutions operating on this level are typically technology transfer centres, clusters, innovation service providers and funding agencies. They can be considered as the tools to turn political decision on innovation into practice. In emerging countries, such institutions are mainly publicly owned. Their main tasks are to serve the stakeholders in order to increase their competitiveness and capability to innovate. Institutions on this level generally do not own separate programmes to fund innovation support measures rather than providing in-kind contributions by training, consultation, conducting applied R&D or product improvements. Nevertheless those institutions are one of the major pillars to improve the innovation capabilities of firms, especially in those countries where public investments are limited.

Meso Level: Programmatic Innovation Support Level

By programmatic innovation support we mean public funding programmes and initiatives used to turn innovation policy into practice. It represents the second important pillar to improve the innovation capabilities of the stakeholders within a NIS. Such programmes can be managed by policy makers or innovation support institutions. Nevertheless, measures on that level need significant public investments.

Micro Level: Innovation Capacity Level

The micro level builds the umbrella for the main practical actors and enablers within a NIS, like SME, entrepreneurs, universities, public or private R&D institutions, innovators or financial facilities.

The different dimensions are influenced by certain determinants. These determinants are of dedicated interest for our analyses since they can be influenced and improved by appropriate measures. In summary, the pattern of determinants on all four levels mostly affects a NIS. Although we use the four levels separately, we acknowledge that there are plenty of interdependencies and links between them. But it seems to be appropriate to consider the levels separately during the analyses of NIS in Jordan. We are also aware that the different determinants can influence a NIS differently. The ideal way for Jordan to improve the outcomes of its NIS is not the same as the best way, for example in Italy. Thus, we did not evaluate the pillar or the respective determinants, although it is obvious that certain determinants are usually of higher relevance than others. At this point we want to point out that a NIS is, of course, also affected by factors coming from outside the country. Within a globalised world, all NIS are affected by external influences. But, since these factors cannot be controlled by national policy makers and actors within a NIS, they will not be considered in the present analysis.

In total we identified a core set of 29 determinants below the three levels, which, by our perspective, are considered worth to be analysed within this study. All of them directly influence the efficiency of a NIS. All determinants can directly be addressed by different measures. Some of them by short-term and with low efforts, others need long periods of time for improvements, combined with significant investments. Improving a certain determinant can have manifolds positive impacts.

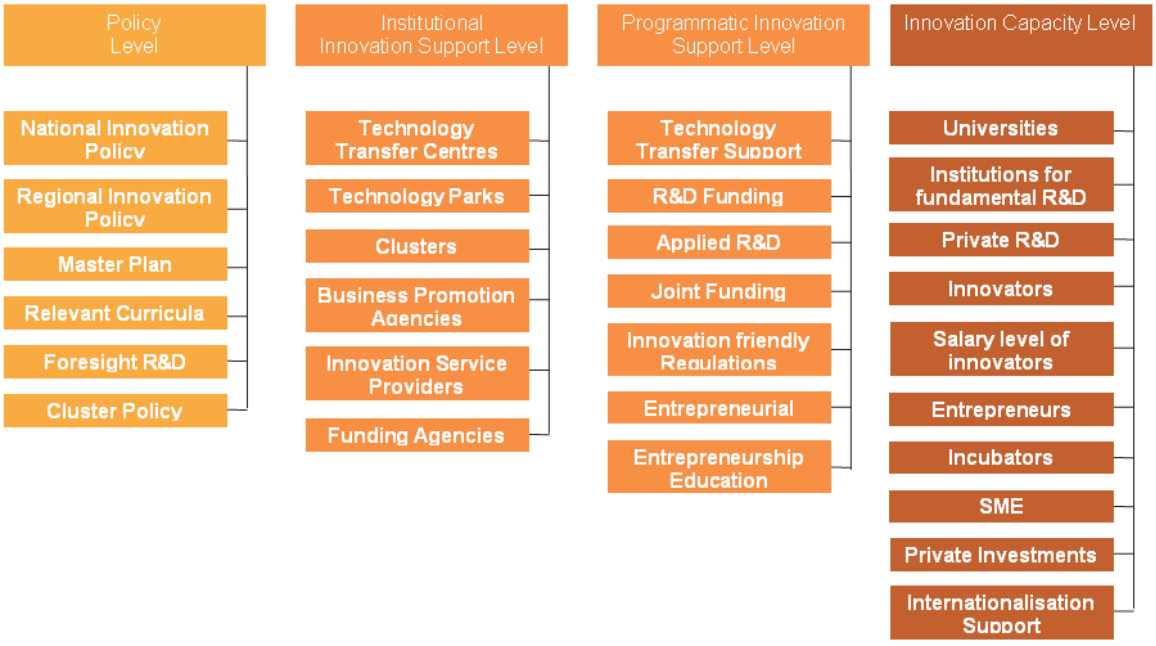


Figure 3 Main determinants of a national innovation system

In order to assess the stage of development of all determinants, we have designed between three and five questions for characterising the 29 determinants accordingly. The onsite assessments were done by national experts as well as by the expert team of VDI/VDE-IT as explained below.

Expert Opinion Survey (EOS)

The model used draws on a wide range of data from the Expert Opinion Survey (EOS). The EOS meets the need for up-to-date and far-reaching data, providing valuable qualitatively information for which hard data sources are scarce or nonexistent. The survey was completed by at least 20 national experts for each country. We asked the experts to provide their opinions on various aspects of innovation and the innovation environment in which they operate. The data gathered thus provide a unique source of insight and a qualitative portrait of each nation's innovation concept and how it compares with the situation in the other country.

The questions in the study follow a structure that asks the interviewees to evaluate, on a scale of 1 to 4, the current conditions of their particular operating innovation environment. At one end of the scale, 1 represents the worst possible operating condition or situation and at the other end of the scale, 4 represents the best (see figure 4).

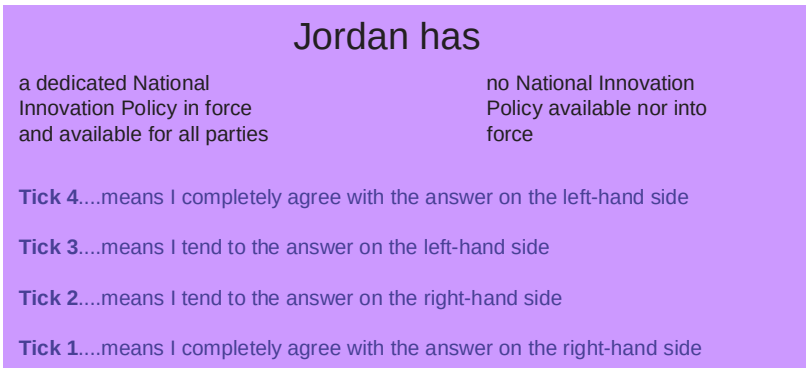


Figure 4 Example for a typical question

The experts for the EOS have jointly been selected by GTZ as well as by VDI/VDE-IT and its local partners in order to cover the complete spectrum of determinants of a national innovation system. This valuable collaboration helped to ensure that the interviews were conducted in a consistent manner across both countries. In addition, our partners supported us in explaining the approach at the national level. Nearly all of these experts have been interviewed by VDI/VDE-IT with support of local partners and most of them have completed the survey after the interview. Those experts, that have not been available during the stay of VDI/VDE-IT in the specific country, have been asked to complete the survey on their own.

The Indicator Approach

Based on the findings of the EOS, we then were in the position to calculate appropriate indicators based on the evaluation of the questions grouped around the one determinant. We also used a scale which was very similar and therefore comparable. When it come to the scale,

- Indicator of "1" represents the determinant at its worst operating condition or situation pointing out that a certain determinant is very poorly developed or not existing. This value was generated if all the questions belonging to that determinant were answered by "1", too.
- Indicator of "2" means, that a certain determinant basically exists and has shown first positive impacts. Nevertheless, there is a strong need to improve the efficiency or functionality.

- Indicator of “3” means, that a certain determinant is matured and has shown positive impact on the performance of a national innovation system over a longer period of time. Nevertheless, there is still room for further improvements towards an excellent performance.
- Indicator of “4” represents the determinant at its best operating condition. It is strongly developed and well operating in practice over a long period. However, some very slight improvements may be still possible

Indicator values above 3 are characteristic for well developed industrial countries where all determinants are established and well functioning, some slightly better than the others. A value between 1.5 and 3 means the determinant already exists and is in the phase of development. Values below 1.5 mean that a specific determinant may exist, but is not really operational in practice or it needs improvement. This is characteristic for those countries that have a quite weak NIS in force.
The following figure gives an example for the chosen approach to visualise the results out of the expert interviews. The green tab marks the individual determined results.

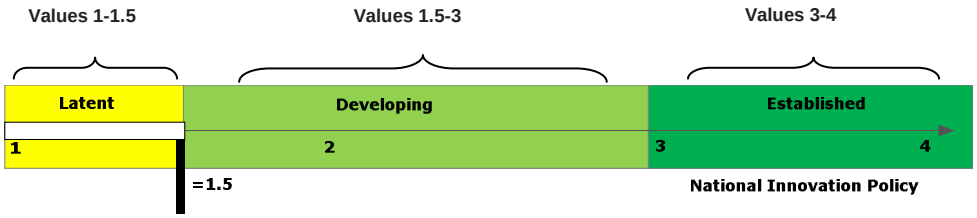


Figure 5 Example for the visualised results out of interviews

Analysis of Potential Improvement

Describing the determinants is important in order to better understand the quality and stage of development of a NIS. If a NIS should be improved, certain determinants must be addressed. It is quite clear that certain determinants are easy to improve, while others are much more complex. For that reason, we grouped the determinants according to their complexity for improvement. We used a portfolio with two different scales. One scale represents the “efforts needed” (how extensive is the needed amount of investment to enhance the performance of the determinant?), the other represents the “expected impact” (what range of improvement can be expected?).

Thus we focused our recommendations on those which do not need a huge amount of effort, in terms of money, training, etc., but have a significant impact on the overall performance of a NIS.

3 Executive Summary

Innovation is the ability to take new ideas and translate them into commercial outcomes by using new processes, products or services in a way that is better and faster than the competition[8]. Certainly innovation cannot be organised by decree. It needs stakeholders and people being active within a NIS and only these people - scientists, researchers, entrepreneurs and their employees, firms, investors, consumers and public authorities — can make a country or region more innovative and competitive. But they do not act in a vacuum. They act with a mindset and in a framework which either discourages or incites them to enter unknown territories, the so called national innovation system.

The main objective of this study was to analyse the Jordanian NIS as well as to identify future supporting measures in order to increase the competitiveness of local SME as well as the innovation capability of Jordanian actors. Special focus was given to a set of recommendations for improving actions, which may be implemented and funded by external donors, like GTZ in the near future (chapter 5).

As far as the analysis of the NIS was concerned, an indicator-based approach was chosen, involving many local experts, who provided GTZ and VDI/VDE-IT with their expert opinions. The so called **ANIS**[9] approach is based on 29 determinants, which, on one side, are very characteristic for the different facets and the level of maturity of the Jordanian NIS and, on the other side, are very much influencing the effectiveness and efficiency of an innovation system. This approach allows to clearly presenting the strengths and weaknesses of the Jordanian NIS in a very transparent way. The determinants were calculated based on interviews with many local experts (Expert Opinion Survey), which then were calculated and transferred on a scale between “1” (not existing or in a very early stage of development) and “4” (well matured). Similar analyses, based on the ANIS approach, were also conducted in different neighbour countries, which allowed to benchmark the maturity of the NIS and its determinants of Egypt, Jordan and Syria (chapter 6).

[8] ‘Creating a National Innovation Framework’, Science Progress, Richard Nedis & Ethan Byler, April 2009.

[9] ANIS – Indicator-based **A**nalysis on **N**ational **I**nnovation **S**ystems, developed by VDI/VDE-IT

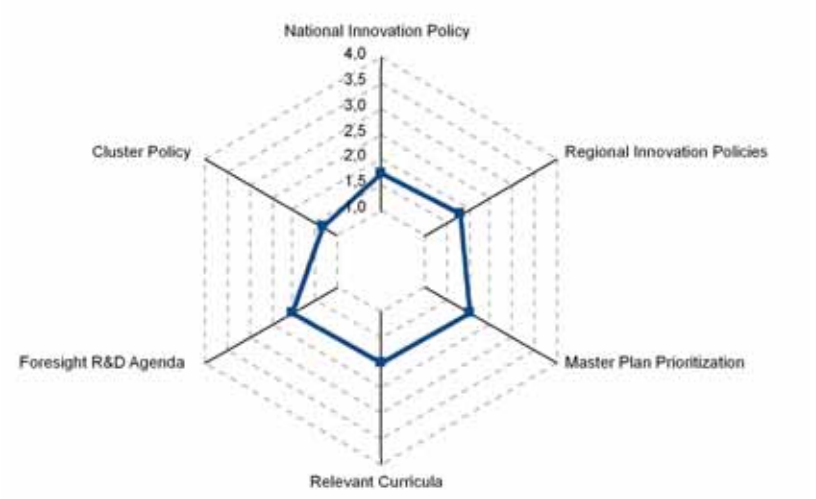


Figure 6 Pattern of the determinants on policy level

An appropriate awareness of the importance and advantages of innovation on the competitiveness of the Jordanian economy exists on all levels. The analysis reveals many examples that good progress were made, especially on **innovation capacity level**. On **policy level** (macro level) the pattern of the six corresponding determinants appears quite symmetric, mostly around the value of 2 (developing phase, s. figure 6). Only in terms of cluster policy, a lower value was gained. According to the local experts involved, cluster policy does still not really exist in Jordan, although there are some promising industrial agglomerations in the pharmaceutical and ICT sector. So far, there is no written national innovation policy available yet. Nevertheless, members of the Jordanian government are well aware of needs for a strong innovation policy. They tend to announce their own innovation programmes without having coordination among the entire government-levels. Although foresight-processes concerning the prioritisation of future technological domains, incl. R&D, are not officially politically supported, many experts from academia and industry in Jordan seem to be somehow involved in such discussions about future technological developments and challenges, whereas these determinants was rated quite high compared to others.

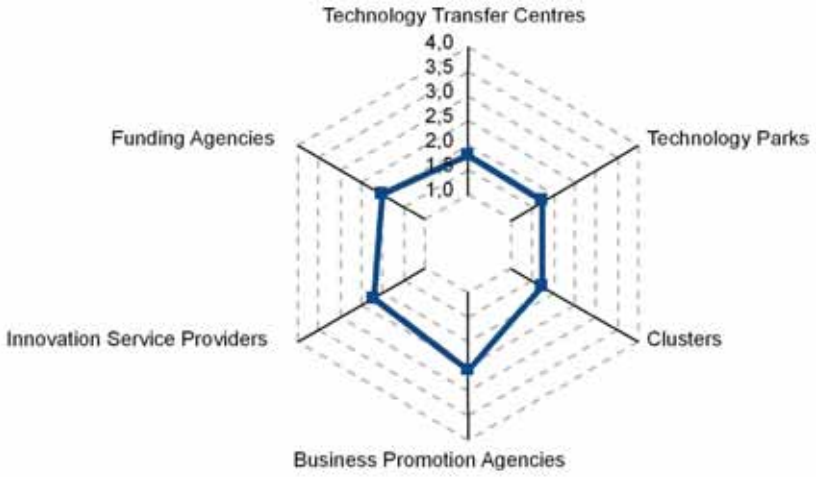


Figure 7 Pattern of the national institutional innovation support level

Assessing the **institutional innovation support level** (meso level) in Jordan, the pattern of the six determinants reveals some interesting characteristics. Business promotion agencies are highly recognised as tools to support firms in certain innovation matters and seem to play a comparable strong role in Jordan. Also innovation service providers are assessed to be further developed than the other institutional innovation supporters (figure 7). Since the Jordanian government, partly supported by external donors, spent significant efforts to support business promotion agencies and certain innovation service providers in the past years, the positive outcomes nowadays become visible, resulting in a comparable high rating of the corresponding determinants. Determinants related to technology transfer centres, technology / science parks and clusters are rated below average, which means that these tools for turning innovation into practice are not as matured as the previously mentioned.

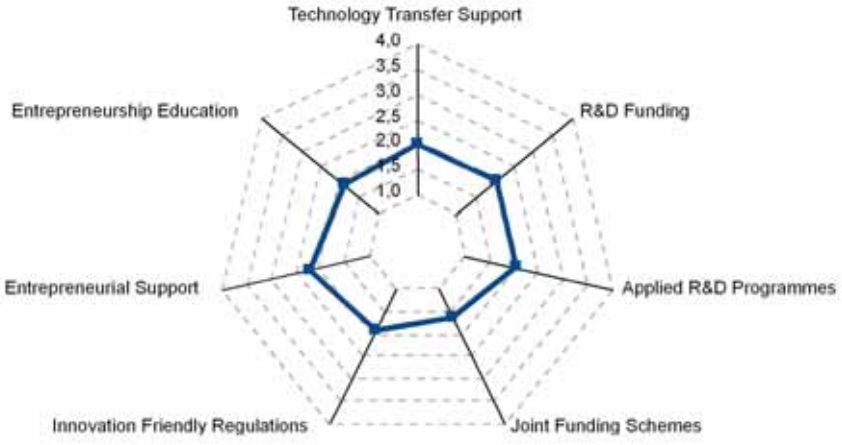


Figure 8 Pattern of the national programmatic innovation support level

The pattern of the seven determinants related to the **programmatic innovation support level** (meso level) yields some interesting characteristics as well (figure 8). The pattern appears to be well balanced, with an average value of 1.9. This means that the determinants are in the developing phase. Highest values can be observed for programmes aiming for entrepreneurial support. Jordan has also started to stimulate technology transfer between researchers and industry by launching appropriate programmes, like Faculties For Factory. It aims to increase the industrial activities to become more competitive as well as to improve the technological infrastructure.

Using public funding programmes to address R&D and certain innovation issues is still a new subject in Jordan. The implementation of competitive funding is at a starting position, but making good progress. As an example, the Jordanian government set up the Fund for Scientific Research in 2007. Lessons learned during the first call are not turned into practice.

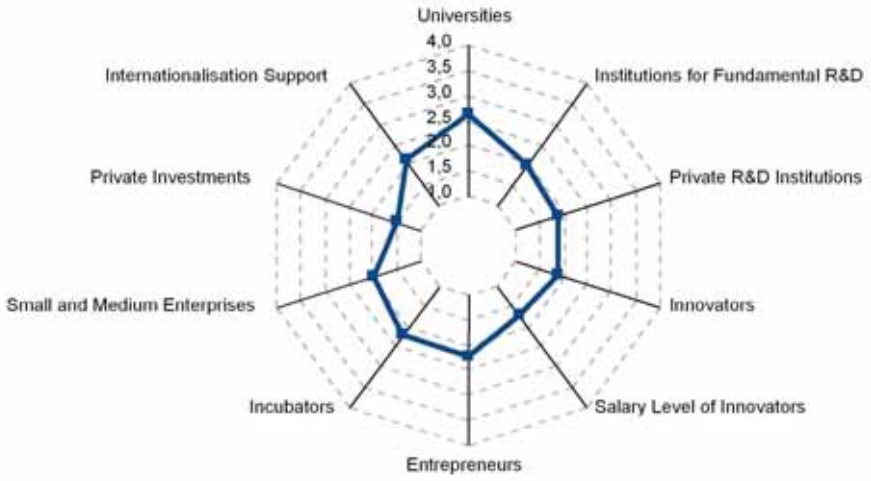


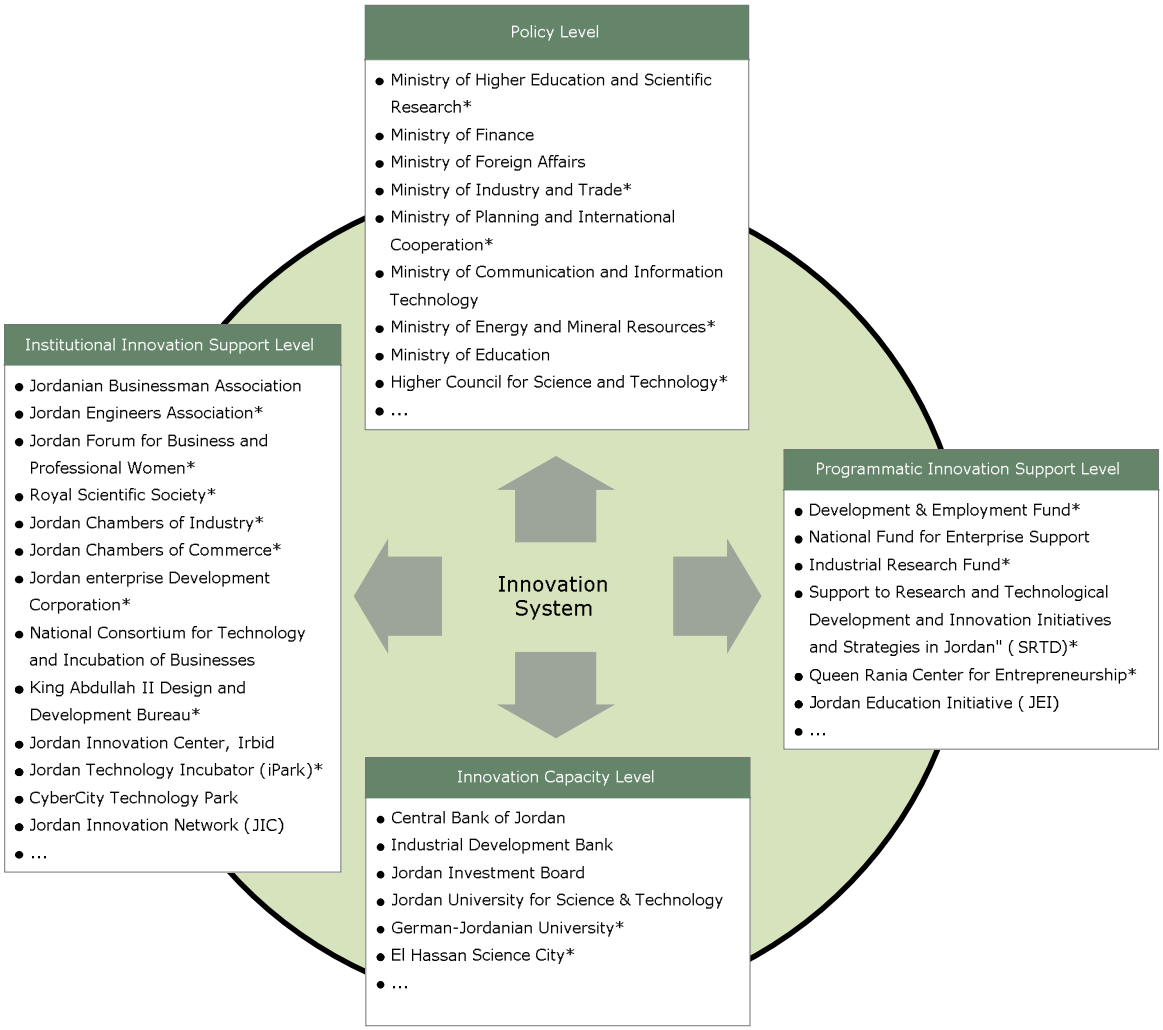
Figure 9 Pattern of the national innovation capacity level

Universities are the determinants that are best developed on the **innovation capacity level** in Jordan according to the findings (figure 9). To enhance the capabilities of researchers to transfer academic results into products and services, there are several promising approaches existing. Entrepreneurs are also considered to be an important group of actors within the NIS in Jordan, since the corresponding determinant was also ranked over average. Most SMEs are currently not in a position to act as drivers for innovation. There are only very few innovative SMEs, which can be considered as high potentials. Private investments in R&D are still an exception in Jordan.

Based on the findings a set of recommendations were identified, how to improve the effectiveness and efficiency of the Jordanian NIS. It is obvious that some determinants of a NIS are easy to improve, others are much more complex. Based on a portfolio analysis especially those determinants were identified and addressed by recommendations, which can, on one hand, be implemented with reasonable efforts and, therefore, are of dedicated interest for third party donors. On the other hand, a successful implementation of these recommendations may have a significant positive impact on the Jordanian NIS in the near future. These recommendations focus on

- Designing and implementing a harmonised National Innovation Strategy, which is the base for all currently ongoing and planned innovation support measures
- Further improvements of the framework conditions for entrepreneurs
- Increasing the capacities of institutional innovation supporters
- Turning existing industrial agglomerations into innovative networks and clusters by setting up and implementing a National Cluster Policy, accompanied by appropriate supporting measures

4 National Innovation System in Jordan



* interviewed partners

Figure 10 The main players in the Jordanian innovation system (besides of the industrial actors)

The Jordanian NIS contains plenty of actors on different levels as shown in figure 10. Even the number and names account for many activities to this respect. In the following chapters we assessed the 29 determinants, grouped according to the three levels of a NIS. This has been conducted based on the Expert Opinion Survey, which involved a significant number of the expert interviews, as well as on the impression of the expert team. This approach led to a whole picture of the NIS Jordan. In the beginning, we grouped the determinants according to the respective levels (macro, meso and micro) and gathered them in one spider graph. Then we described each single determinant, starting with a short summary, followed by a more detailed description and a short comment, whether potential improvement may by reasonable taking the limited resources especially for third party donors into account.

4.1 Policy Level

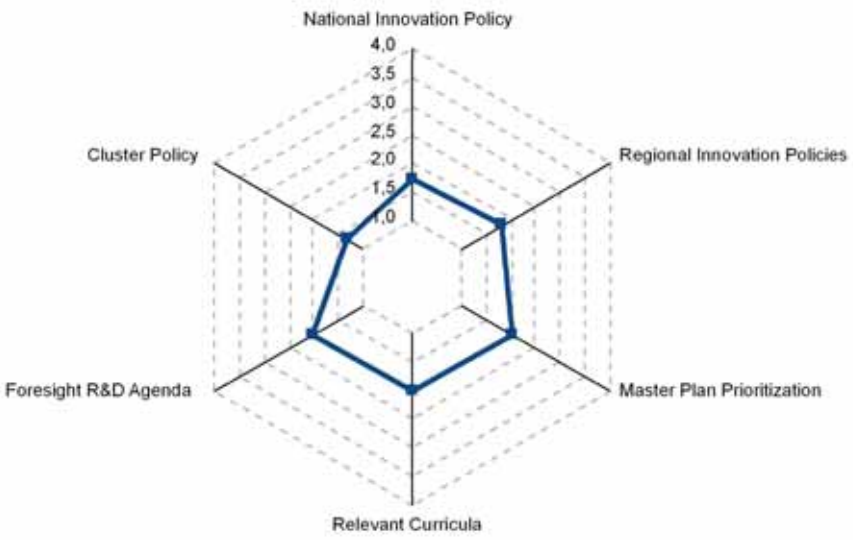
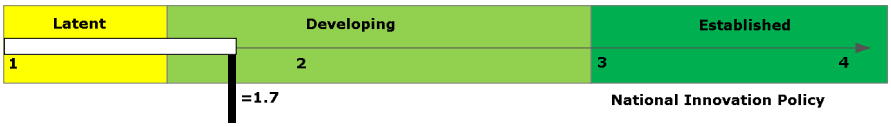


Figure 11 Indicator analysis of innovation policy level - Jordan

Figure 11 summarises the six determinants of the policy level underlining the current phase of the strategy-processes to develop innovation related policies in Jordan. The pattern appears well balanced; the values are all laying in a comparable order. Most of the scores reveal that the determinants belong to the developing phase. Innovation is a strong topic on the policy agenda, but still more related to R&D; policies related to cluster development are not yet in a strong focus so far. In the following, the determinants are described and assessed more detailed.

Findings: National Innovation Policy



An appropriate awareness of needs for innovation policy exists on all levels of the Jordanian NIS. There is no written national innovation policy available yet.

Members of the Jordan government are well aware of needs for a strong innovation policy. They tend to announce their own innovation programmes without having coordination among the entire government-levels. During the interviews with experts from the policy level, the national innovation policy was often judged as being established whereas experts on other levels mainly judged national innovation policy at least as latent and even often as not existing at all. Asking for reasons, it was mentioned, that ministers change rather often and with them, innovation programmes are changed as well. Most experts miss a stable long-term-commitment to innovation support.

The specific elements of the NIS should be better coordinated according to the EOS. It is remarkable, that nearly all experts mentioned this without having any idea, how to change. On the other hand, a considerable number of persons in organisations on all levels are highly engaged in the

innovation processes in Jordan. But they do not feel significantly supported there. They are struggling on their own, competing against other innovators for funding of projects, they want to realise. Furthermore, this kind of competition tends to have a negative impact on the coordination.

All these personal innovators define their own objectives and act this way. Often they are not even aware, of having an active role within the NIS.

A Jordanian national innovation policy can be implemented with little efforts, but high impact. The strategy should be written, stable but adaptable if necessary, independent of persons holding offices or ministries and broadly published. A strong involvement of many important actors from policy, industry and academia is strongly recommended. Funding and legislation must allow innovation processes on all levels to have planning security.

Findings: Regional Innovation Policy



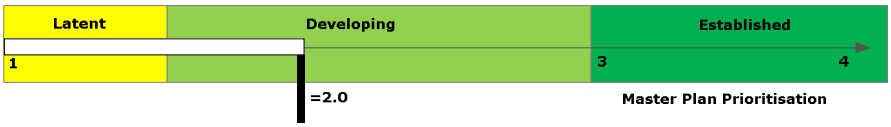
A regional innovation policy is not formulated in writing. But there is potential of developing specific branches and regions in Jordan. Implementing regional innovation policies according to the local demand could be a promising tool to use the existent potentials.

“Inside Amman” and “outside” was mainly distinguished during interviews. This is replicable, considering that one fifth of the inhabitants live in Amman and desert covers most of the countries area. Experts have partly been aware of potentials, regional innovation policies can have. In Zarqa and Irbid, one third of the Jordanian population is living and in Zarqa, about half of the Jordan industry is situated. Both cities are situated in the north of Jordan. It is obvious, that regional innovation policies for Amman, Zarqa-Irbid region as well as for the rest of the country are strongly needed in order to implement regional-specific and demand oriented innovation policies.

Besides, Jordan has established "Qualified Industrial Zones" (QIZ), where laws regarding taxes, customs and other economic issues are suspended. In these zones, foreign companies find suitable economic frameworks. This is done mainly by the influence of the UN to ease conditions for Palestine refugees. These QIZs could become an important part of the NIS, especially supported by regional innovation policy. But nowadays experts do not see any significant influence of the QIZs on the Jordanian economics as most of the employees come from abroad and technology transfer is nearly not affected.

Regional innovation policies and appropriate actions are mainly based on deeper analyses of the strengths, potential and demands of the regions in terms of innovation. To assess the potential and the perspectives of different regions in Jordan, a deeper, well structured analysis seems to be promising. Each region requires a tailor-made approach, designed to the real needs of the partner who could be responsible for the implementation. The expected impact of regional innovation related activities seems valuable, the mandatory effort will be low.

Findings: Master Plan Prioritisation

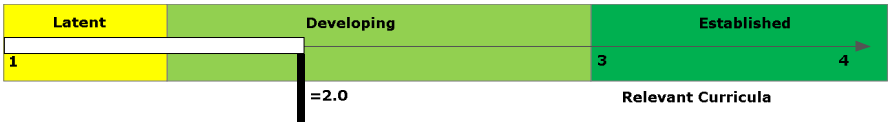


Jordan has implemented a process of formulating national research strategies as part of innovation master planning. The corresponding five-year plans need to be further developed and consequently turned into practice.

On behalf of the government, the Higher Council for Science and Technology (HCST) develops national research strategies for five-years-periods. The next one, covering the years 2011 to 2015 is under discussion. According to the interviewed experts, there are some concerns, that these strategies will be adequately regarded by Jordan policy makers - even though all the ministers are shareholders of HCST and involved in the development of the strategy. Attention must be given to assure that strategies developed by the HCST have an impact on Jordan's research and innovation policy. Many experts outside the government which are highly engaged in innovation expressed their wishes to become actively involved in the process of innovation strategy developing, of the HCTS. This would assure high practice orientation and common acceptance on all levels of the Jordanian NIS.

The further development of a NIS to define practical and mandatory objectives for research and innovation could be implemented with low effort and high impact. Corresponding master plans in order to turn such policies into practice also do not cause too much effort, but can provide a significant impact on regional level. The Jordanian government should decide to support the process by external moderation, regular assessment of distinguished issues and continuous evaluation.

Findings: Relevant Curricula



Jordan's government has recognised the importance of up-to-date and demand oriented training and education. Curricula of public and even private universities are existent, but are not considered to be fully in line with the industrial demand. Mainly theoretical knowledge is addressed by the current curricula.

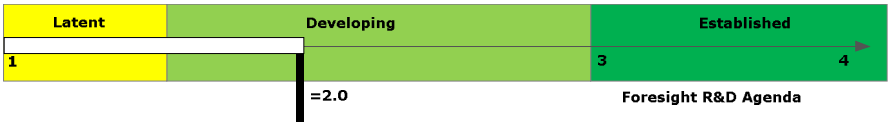
Some experts are informed about the considerable efforts of the Jordan government to develop vocational training with support of international financing. According to the EOS, the Jordanian

curricula for education at schools and universities as well as vocational training do not fully meet the current needs. This finding is backed by EJADA, which has reported[10], *that teaching and training focuses on facts that have to be memorised instead of teaching methodologies to solve problems. Public schools often lack the minimum standard of infrastructure and curricula to provide an environment conducive to innovations. Vocational training does not match the needs of industry. Teaching and training are not at all demand driven.*

Some concerns have been expressed, that time-frame for enhancement of vocational training system and curricula may be too long to fit the needs of industry and trade with respect to speed of changes in competitiveness of the Jordan economy. Besides, comparable efforts on basic education at schools and high education at universities are missed in many experts' perspective.

The enhancement of curricula on all levels of education and vocational training could be done with reasonable effort but high impact on innovation. Implementation of fast-acting measures, like special courses with curricula on international standards, allowing international certificates, should be considered. These efforts must follow an assessment of the major and urgent needs of the Jordanian economy.

Findings: Foresight R&D Agenda



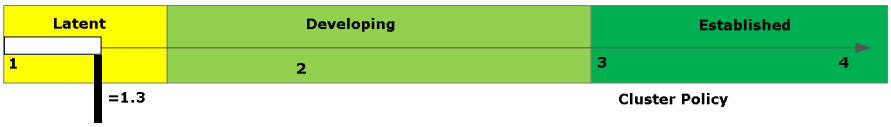
As a consequence of a missing NIS, Jordan has not fully implemented any special foresight-process for the development of future oriented research strategies or to identify future technological domains of high priority.

A national foresight process for R&D in Jordan can be seen as part of the national research strategy developed by HCST, but must include a wider range of experts of all groups of the society. Many interviewees are aware of such needs. We have gained the impression that such discussions are ongoing between scientists and innovation practitioners. Thus, foresight is scored as developing as and higher than in other countries in the region.

Missing R&D foresight processes is not one of the main problems of Jordan. It could be implemented with rather low effort and could significantly raise awareness of innovation and form communities for long-term cooperation. It also could contribute to identify promising technological domains and country-specific challenges which then could be addressed by the national innovation system.

[10] David Tee, A Practical Guide to Wealth Creation, Proposals for an Innovation Policy for Jordan, EJADA, 2006

Findings: Cluster Policy



Jordan has started to set up and implement a national cluster policy. This process is in a very early stage, but first would-be clusters have become visible. Further efforts are needed to increase the impact and efficiency of a national cluster policy on the innovation capability and competitiveness.

In the beginning of this century, Jordan decided to develop a strong cluster driven policy to develop its pharmaceutical industry[11]. This is not a surprise since the pharmaceutical industry thought to be still competitive especially in comparison to neighbouring countries. Jordan's IT sector can also be used to demonstrate a relatively young but successful sector that has capitalised on the cluster approach. In that case the cluster emergence was mainly pushed by the private sector[12]. The role of national cluster policies in economic growth and competitiveness is not doubted in developed countries. Clusters are useful frameworks to coordinate strategies and reduce complexity.

Cluster policies often result in specific cluster initiatives. In consequence, cluster initiatives can be understood as “organised efforts to increase growth and competitiveness of clusters within a region, involving clusters firms, government and/or the research community. They often need a political support that balances creating and sustaining of innovative clusters. Nevertheless, the awareness that the Jordanian government pays attention to this topic seems to be quite low as many interviewees do not see items of a national cluster policy in practice.

Jordan should develop a long-term cluster policy since there are potentials in several technological domains. This must be based on comprehensive assessment of existing economical sectors covering competitiveness and future potential. This assessment should be done by external agencies, involving and training future Jordanian cluster experts to sustain cluster policy in Jordan.

[11] http://www.competitiveness.org/article/articleview/63/1/28

[12] According to the Jordan Times, 2004

4.2 Institutional Innovation Support Level

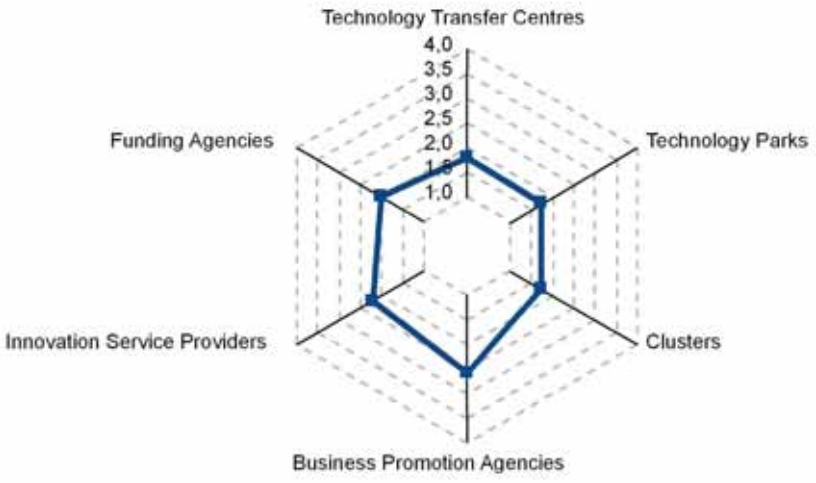
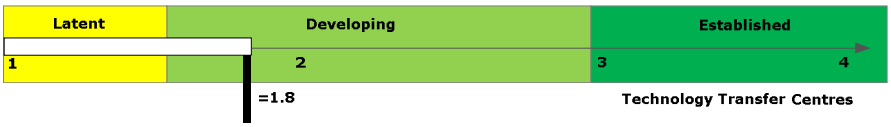


Figure 12 Indicator analysis on institutional innovation support level - Jordan

The expert opinion marks with the six determinants of the institutional innovation support level a realistic summary of the ongoing activities in this field in Jordan. The average of all determinants related to this level is slightly higher than that of the policy level. The business promotion agencies and the service providers in the field of innovation are ranked comparably high and on a promising way of development. Technology transfer centres and funding agencies seems not to be as far developed as business promotion agencies and service providers.

Findings: Technology Transfer Centres



There is a commitment for implementation of Technology Transfer Centres in Jordan. As an outcome of it, several Technology Transfer Centres exist in Jordan, mainly connected to universities. There seems to be a good potential that Technology Transfer Centres can play a more active role in the Jordan NIS, provided they receive a stronger political and financial backing than in the past.

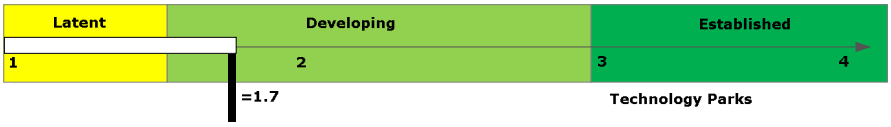
Researchers and industry are not in the position to develop competitive technologies for international markets without institutional innovation providers, like technology transfer centres. This is not only valid for Jordan, but rather for all emerging countries world-wide. As far as Jordan is concerned, the regulations for some universities set out clear mechanism in which scientific centres for

technology transfer can be established[13]. Examples of existing centres are Hamdi Mango Centre for Scientific Research and National Energy Research Centre at the Royal Scientific Society as well as Princes Haya Biotechnology Centre and Pharmaceutical Research Centre at Jordan University of Science and Technology. USAID Jordan Economic Development Programme (SABEQ) has done an assessment on technology transfer in Jordan and stated that these centres would be capable to contribute to technology transfer, but can't do it sufficiently in reality, due the lack of funding[14]. Sufficient financial and personal resources have been identified as one of the major success factors for technology transfer centres in emerging countries.[15]

As a consequence technology transfer centres in Jordan are not recognised as a major actor of the NIS, although they would have the potential.

Based on recommendations for implementation of technology transfer at Jordanian universities, given by SABEQ, technology transfer should become more important in Jordan. The potential and starting position is quite good since there is a certain number existing with close connection to the industry. However, readiness for public investments is the key to improve the performance of the Jordanian technology transfer centres. Without increased funding, technology centres will remain as they are. Nevertheless, public investments in such centres will have a very positive impact on the creating innovation products and technologies and will considerably support Jordanian SME to increase their competitiveness.

Findings: Technology Parks



Two technology parks, focused on ICT, have been built with the support of international bankrollers. So far their activities seem to be quite low.

UNIDO[16] reports about the corresponding technology parks, focussing on ICT. infoDev, a programme mainly funded by World Bank, reports in 2008[17] as well, that the parks have been constructed. Both reports can not verify how active they are, when they started working or whether they were still under development. One of these parks, CyberCity, is located as a new Qualified Industrial Zone, purposefully to support the development of the IT industry, near Irbid, 85km north

[13] R&D Assessment at Jordanian Universities: Encouraging Transfer of Technology to Innovation Stage, USAID, 2009

[14] ibidem

[15] Meier zu Köcker, G; Hahn, P, The National Innovation System in Egypt – An exploratory study of Egyptian Technology Transfer Centres, financed by GTZ, www.iit-berlin.de/veroeffentlichungen, 2008

[16] http://www.unido.org/index.php?id=o26470

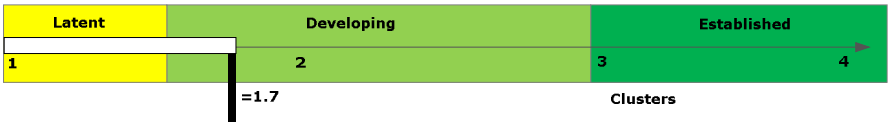
[17] International Good Practice for Establishment of Sustainable IT Parks

of Amman. According to the infoDev report, one major obstacle may have been that the park was planned far outside of existing IT clusters without an incentive for companies to move there.

There might be an idea of developing Hassan Science City (HSC) to become a technology park. So far, HSC is working as a campus on which several organisations (Royal Science Society, HCST, Queen Rania Centre for Entrepreneurship...), universities and incubators are located. These organisations are still waiting for a law that makes HSC a legal body. Before, they can not act or contract on their own.

Technology parks seem to be some steps ahead of the potentials and needs of Jordan innovative industry and commerce. Due to the high financial efforts, they should be seen as a long-term perspective, provided further investments are needed. It was not the scope of this study to assess, why the existent technology parks are not considered to be as effective as they might be. Key questions are, whether they have a dedicated strategy into force, how to attract companies and knowledge providers, which kind of services are needed for the tenants and how to increase national and international visibility. Support in this respect may be a field of further advice. Establishing more incubators and strengthening existing technology transfer seems to be a more effective approach than setting up more technology parks.

Findings: Clusters



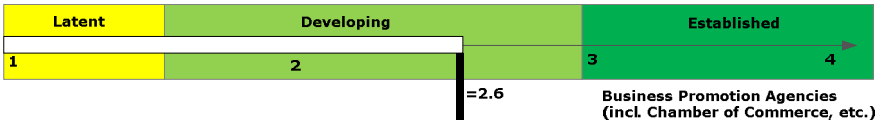
There are some regional networks or clusters existing in Jordan. They have been emerged according to a bottom-up approach and are mainly industrial driven. There may be nucleus for further cluster development in Jordan, but maybe these future networks and clusters need political support.

In Jordan one can find some "natural clusters" or regions where certain branches are concentrated and acting in networks. For instance, clusters for pharmaceutical or ICT industry have been mentioned by our experts. These clusters, also bottom-up clusters, are characterised in that they emerged by a gathering of industrial and scientific partners to intensify the mutual cooperation in order to gain competitive advantages for their daily business[18]. Of course, there were different reasons for setting up such a kind of clusters in the pharmaceutical or ICT. The co-operation characteristics within these clusters are typically decentralised. Although in most cases the setting up of clusters in emerging countries is often stimulated by public authorities, it can be understood as a very positive sign that these two networks or clusters emerged without any public intervention.

[18] Meier zu Köcker, Gerd (2008), Clusters in Germany, www.iit-berlin.de/veroeffentlichungen

Clusters for renewable energy, pharmacy and ICT could be further strengthened with low effort, but high impact. Key success factor for the cluster performance and the increased innovation capability and competitiveness of the cluster firms are demand oriented and tailor-made services offered by a cluster coordinating body. This may be the focus of future interventions and support.

Findings: Business Promotion Agencies (incl. Chambers of Commerce, etc.)



Business promotion agencies are very strong in Jordan.

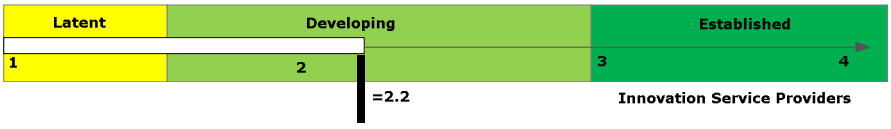
Chambers for Industry and Commerce are considered to be very strong according to our EOS. No business is possible without membership unless no rooms are rented or bought (freelancers working at home). Chambers offer full support for all members covering administration issues as well as training programmes for business (management, legal affairs and financial affairs). Acceptance is sometimes low, as in daily life, the benefit of the chambers is not recognised. But the one-stop-shops for solving administration problems and the training programmes are well accepted.

For each professional group, there is a kind of "union" of which they have to be a member. For engineers, membership in Jordan Engineers Association (JEA) is mandatory. JEA cares about personal affairs (fighting for salaries, keeping pension funds, credits for buying houses) and has a great influence on politics. JEA is involved in nearly every construction project, as nothing can be build without being certified by JEA.

The responsibility of the unions is regulated by law. If they could decide on their own, they would engage much more in innovations. During discussions JEA for instance stated, that they would immediately invest in incubators, if they were allowed to. Unions and chambers of industry and commerce know best about the needs of the economy. They could play an important role in the NIS.

Jordan should take advantage of the engagement available in the professional unions just by changing laws and regulations towards more autonomy. This can be done with very low effort but very high impact on innovation.

Findings: Innovation Service Providers



A lot of organisations and persons are acting as innovation providers. Some of them are considered to be very strong. Their impact on the NIS could be further increased by better coordination and programmatic support (e. g. funding).

Many experts stated that innovation service is very successful, provided by many different actors. There is no lack of business promotion agencies or service providers that could deal with innovation support[19]. Despite of this positive finding, no mid-term planning can be established due to unstable funding. It is a "living from hand to mouth" for all actors as no one has any knowledge about the next-year-budget. Political and financial aids for these supporters are available, but only by personal convincement of funds or organisations. Increased budgets for longer periods embedded within a programmatic framework could considerably improve effectiveness and efficiency as well as mobilise even more resources. A better coordination of the work of the several different innovation service providers would help as well.

Further improvements may be gained to lower administrative hurdles. It is often reported that, despite all the support for entrepreneurs (e.g. in one-stop-shops) administrative issues are still time-consuming and time for business-to-market is unnecessary long. But significant progress has been achieved during recent years.[20]

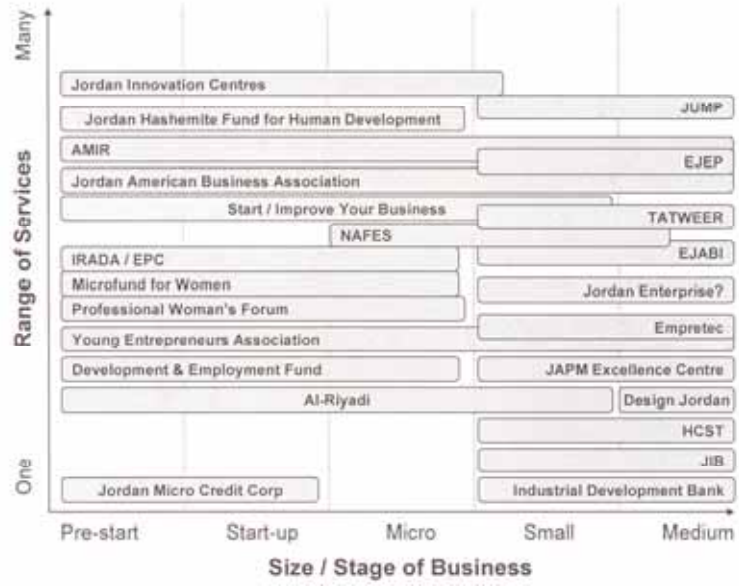


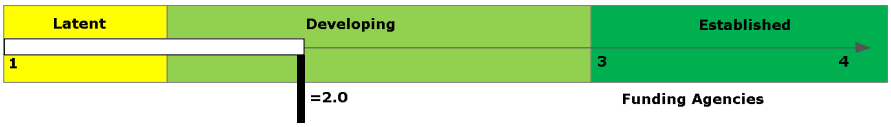
Figure 13 Situation of Jordanian innovation stakeholders

The Jordanian government should give stable political background and funding to innovation service providers. Significant achievements with comparable low efforts are expected as the infrastructure and the actors already exist. With small budgets, for management and entrepreneurship training for example, high impact can be expected. Setting up clear programmatic framework conditions would help as well.

[19] David Tee, A Practical Guide to Wealth Creation, Proposals for an Innovation Policy for Jordan, EJADA 2006

[20] Report on the Implementation of the Euro-Mediterranean Charter for Enterprise, European Commission, 2008

Findings: Funding (and Implementation) Agencies



The administration of funding especially for R&D belongs to the Jordan government.

There is only one special R&D fund, administrated directly by the government (see chapter "Funding"). Administration by a board of directors seems to be rather complex with respect to the available budget and the transparency needed. In the future, Jordan should discuss the establishment of a special funding agency that could, for example, directly support applicants, manage evaluation of outlines and proposals fast and on a professional background and care for the publication of the results and success-stories.

The implementation of a special funding agency is not yet the major need for Jordan especially with respect to the available budget and the efforts that have to be conducted. Once Jordan is setting up and implementing more public funded programmes in the field of science, technology and innovation, having an independent funding agency into force may be beneficial.

4.3 Programmatic Innovation Support Level

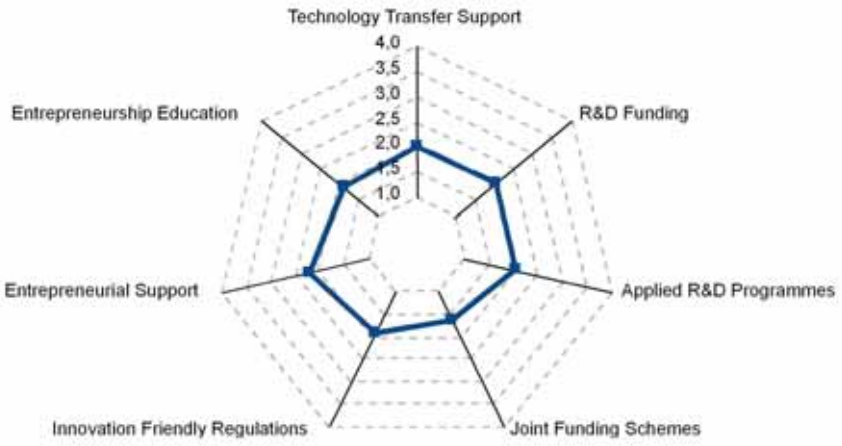
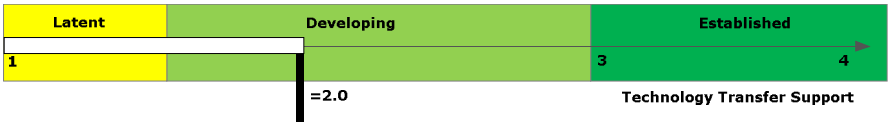


Figure 14 Indicator analysis on programmatic innovation support level - Jordan

The spider graph about the scores across the seven determinants shows, that innovation support is in a developing state. The pattern appears to be well balanced, most values lay around 2.0. This means that the determinants are in the developing phase. Highest values can be observed for entrepreneurial support.

Findings: Technology Transfer Support

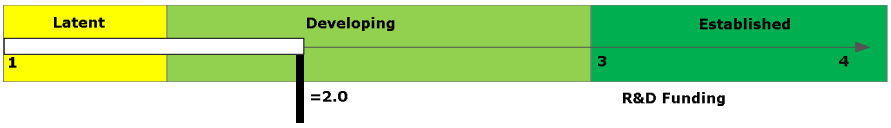


Jordan has started to stimulate technology transfer between researchers and industry by launching a programme called Faculty For Factory. It aims to increase the industrial activities to become more competitive as well as to improve the technological infrastructure.

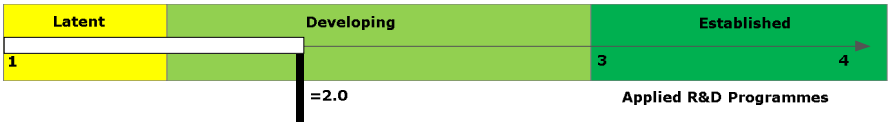
Jordan has started one programme to raise awareness on technology transfer. It is called Faculty For Factories (FFF), run by the Jordan Chamber of Industry (JCI). The aim of the project is to bring academic staff and industry together to solve special industrial problems. Barriers addressed by this programme are on both sides: industry does not believe in academic staff as they are only theoretically oriented and academic staff is not interested as there is nearly no incentive. Nevertheless, 120 faculty members have been involved during 7 programme circles. There is no focus on innovation; the main objective of the programme is just the raising of awareness for needs of applied research and cooperation. According to the Jordan Chamber of Industry, projects funded by FFF are successful for the partners, but awareness in the research and industrial sector is still rather low. JIC still has to “push” partners together as there is rather no demand by industrial or research partners. On the other hand, most of the experts know the programme and mentioned it during the interviews. Further capacities may be stimulated, e. g. by providing incentive for professors and universities to do applied research or become more engaged in technology transfer.

To enhance technology transfer between research and industry is one of the main challenges in Jordan. The programme FFF seems to be a good fundament as it has shown significant success. Setting-up more comparable programmes will mobilise more capacities to innovate, but this needs the readiness of policy makers to invest public money. Corresponding efforts and impacts are expected to be very high.

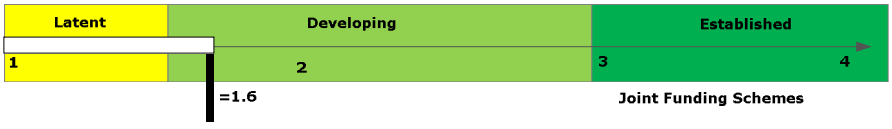
Findings: R&D Funding



Findings: Applied R&D Funding



Findings: Joint Funding Schemes



The finding related to the three determinants regarding R&D Funding are grouped and jointly presented.

Using public funding programmes to address certain innovation issues is still a new subject in Jordan. The implementation of competitive funding is at a starting position. Nevertheless, there seems to be no alternative, since competitive funding is one of the most successful tools world-wide to initiate innovations.

In 2007, Jordan started a Fund for Scientific Research. The budget is defined by law as 1 % of the net profits of all public limited companies (app. 10 million JD/year). The first call for proposals has just been started and experts from the Ministry of Higher Education have to acknowledge that obviously there are some "barriers" for applicants. First of all, researchers are not used to apply for funding in a competitive and formal way. Applicants have to send an outline that is evaluated and to send a proposal after request. This two-step-process, which is quite usual in competitive funding all over the world, is often not understood by Jordanian researchers. At the time of the interview, nearly no proposal was given by industrial researchers in cooperation with universities.

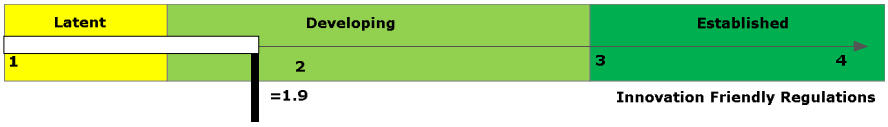
As competitive funding is a rather new but important process in Jordan, regulations should be as easy as possible. Administration of the fund and evaluation of the outlines and proposals seem to be "overloaded" and some special rules, as for instance about returns on property rights revenues are barriers that could be avoided at the moment. SABEQ reports a lot of details about the administration of the Fund for Scientific Research and the regulations[21]. They give several recommendations to decrease the barriers for applicants.

Nearly no public funded programme on **applied R&D** is available today. One exception is a special programme for graduating students, funded by King Abdullah II Design and Development Bureau. Students can propose projects with respect to security and defence. The (20) best of them are funded and the results are published. There is a strong demand for applied research by the Ministry of Defence - but even this demand is not picked up by universities and politics. The FFF programme, mentioned before, seems to be the only programme addressing **collaborative R&D**.

An easy option to improve the efficiency of public funded R&D programmes is related to the just started the Fund for Scientific Research. Application and administrative procedures should be changed in order to avoid unnecessary barriers for researchers and the evaluation process should be made more transparent and linked to the national research strategy formulated by HCST. This would lead to a high impact with low effort. Here, there are many experiences existing from funding agencies in Europe, which can contribute. Besides, it is urgent to increase the budgets for public funded programmes and make them stable for the future. Different R&D programmes should address different technological domains or public challenges. Applied R&D should come into focus of the Fund for Scientific Research, based on industrial needs.

[21] R&D Assessment at Jordanian Universities: Encouraging Transfer of Technology to Innovation Stage, USAID, 2009

Findings: Innovation friendly Regulations



Although innovation is already a topic in the Jordanian society, the enhancement of the innovative culture still needs public intervention. The responsibility is so far shared among many ministries and intermediaries without a clear structure of coordination. Regulations to enable and to support innovations are still missing.

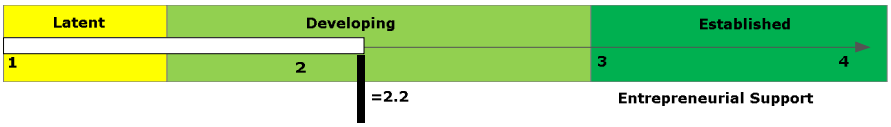
Following most experts, positive regulations for the support of innovation are missing. EJADA published several ideas for such regulations and likewise made proposals for reformation of negative regulations[22]. These recommendations have been confirmed through the interviews with the following exceptions:

Custom affairs have been mentioned to be a barrier for entrepreneurs who want to import products, as long as these products are not clearly destined for industry. It sometimes takes a very long time until custom authorities decide on the duty that has to be paid. This could make business plans for entrepreneurs unreliable.

A mandatory requirement for professionals to belong to professional associations (like JEA for engineers) and for businesses to belong to either a chamber of commerce or a chamber of industry, was only very occasionally discussed as a barrier for innovations. On the contrary, these chambers and associations could be a suitable fundament for innovation support (see chapter "Business Promotion Agencies").

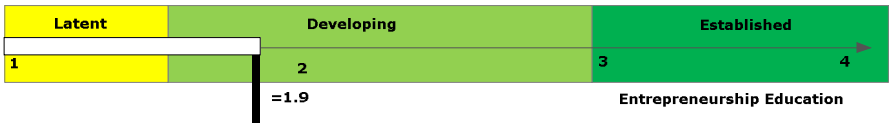
It should be a national task for the government to convince all actors within the NIS that they have to orient all activities on the consequences for innovation. EJADA recommendations are a good fundament for the way to innovation friendly regulations.

Findings: Entrepreneurial Support



[22] David Tee, A Practical Guide to Wealth Creation, Proposals for an Innovation Policy for Jordan, EJADA 2006

Findings: Entrepreneurship Education



The findings of the two determinants regarding entrepreneurship will be discussed together.

There are some effective activities of supporting entrepreneurship into force. The positive impact generated from these activities should be understood as good opportunity for further support. Stable political and financial framework conditions are essential.

There are several activities as for instance in the Queen Rania Centre for Entrepreneurship, or support for business women by the Jordan Forum for Business & Professional Women. These activities are scored to be very effective, but they are driven by highly engaged persons, fighting for financial support most of the time. All interviewed experts expressed their concerns about instability of funding and missed the chance for running mid-term projects that can prove success. For example, there are several ideas for new businesses for women (female craftsman for maintenance at home), but due to the lack of money, chances for implementation of supporting projects are low.

The stigma of failure after foundation of a company or becoming self-employed is still a big problem even though some discussion partners mentioned that it is decreasing. In case of failure, the whole family is affected (due to guarantees, see chapter “Private Investments”). This is a great barrier in the Arabic culture.

Entrepreneurship support should come into focus of the Jordanian policy, since there are good results gained so far. We noticed a positive impact as a consequence of previous actions. Nevertheless, since entrepreneurs are considered to act as key drivers for innovation, the number of entrepreneurs in Jordan should be increased. Therefore it is well suited that Syria participate with own activities in the yearly Global Entrepreneurship Week.

Further support measures are considered to be essential to further improve the entrepreneurial surrounding. Stable funding conditions should be provided to existing and new actors in order to enable mid-term projects. Implementing smart programme design features would help as well.

4.4 Innovation Capacity Level

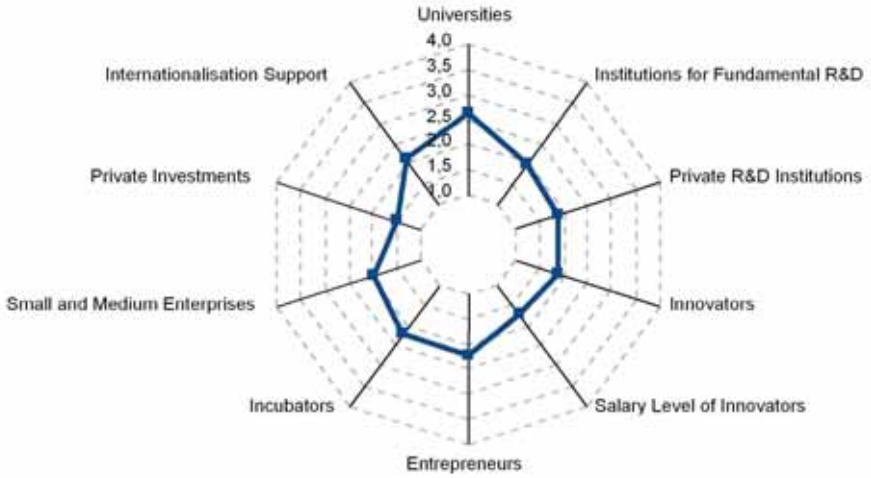
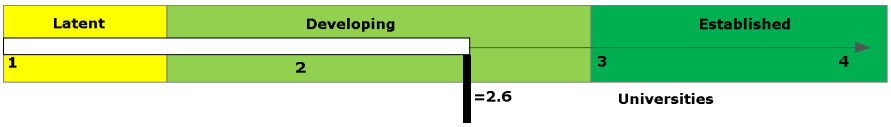


Figure 15 Indicator analysis on innovation capacity level - Jordan

The ten determinants to evaluate the innovation capacity support provide a good picture of the level of maturity. It also reveals the strengths as well as the weaknesses of the innovation-related actors. Universities, SME, entrepreneurs, innovators or incubators seem to be quite strong and on a promising way. Even the capacities regarding R&D are not that weak in Jordan.

Findings: Universities



The number of public and private universities is very high in Jordan and most relevant technological domains are covered. Thus, the respective determinant was ranked comparably high. So far, most of the universities are focussing their activities on teaching.

According to the findings based on the EOS, the national landscape of universities appears to be complete. Some experts expressed the opinion that the number of universities is considered to be too high. Currently, there are 8 public and 13 private universities in Jordan (the country has about 5 million inhabitants).

These universities are mainly focused on theoretical teaching. Research activities play a minor role, especially not in applied research. Teaching quality is judged to be good, compared to other surrounding Arabic countries. As a result, many graduates are leaving Jordan for work in neighbouring countries. This brain drain may become an issue in the future, if the working conditions in surrounding countries remain better than in Jordan. The performance of the academic staff is measured on a regular base (focused mainly at (theoretical) publications). Therefore, incentive to do applied

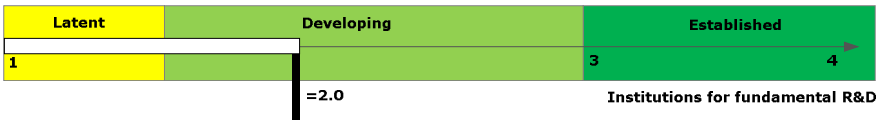
research, especially in collaboration with enterprises, remains low. Engagement of students for innovation topic is still low, as most of them are focused on jobs in government or administration.

The acceptance of university degrees in the Jordanian society is very high. Students or their parents spend a lot of efforts to obtain university places. The chances depend on the achieved grade at school. Beside, private universities are very expensive with respect to normal income of families. Either governmental jobs in Jordan or industrial jobs abroad are in the focus of the most students.

Nearly all experts claimed the existing barriers between universities and enterprises. Industry is not willing to spend money for research at universities. EJADA[23] has reported several recommendations that should help to overcome this barrier on a mid-term run. These cover inexpensive activities like awareness raising, adapting regulations for property rights and allowing academic staff to work in industry as well.

The restructuring of the university system is a cost intensive process due to necessary investment in research infrastructure. This has to be accompanied by more incentives to run R&D on behalf of the industry. Besides that, EJADA´s recommendations for inexpensive activities can be a good fundament for future development. We are fully in line with these recommendations, but do not see this determinant in the focus of third party interventions.

Findings: Institutions for fundamental R&D



Research in Jordan is done in public institutions mainly. Industrial research activities are existent, whereas most activities are unknown to public authorities and other interested parties.

Public institutions for fundamental R&D inside and outside of universities are mainly focused on theoretical work. Nevertheless, private research in some industrial branches exists. For instance in the pharmaceutical industry, some activities have been reported. Since such activities are often not supported by public funds, there is no reliable information available.

An evaluation of the industrial needs for research and existing activities would be the first step to further develop the research landscape in Jordan. This must be accompanied by primary fields of actions, usually defined in the NIS. Significant investments in the Jordanian R&D infrastructure are mandatory, accompanied by public funding. Respective efforts are very high, whereas the impact lasts many years before becoming visible.

[23] David Tee, A Practical Guide to Wealth Creation, Proposals for an Innovation Policy for Jordan, EJADA 2006

Findings: Private R&D Institutions



Private R&D on the institutional level is very seldom in Jordan and only existing at private universities.

Private universities can be seen as private R&D institutions in some cases. Other institutions have not been mentioned by the interviewees. Most of them operate quite theoretically oriented, resulting in a weak impact on the Jordanian NIS. At the German-Jordan-University it became clear that most of the students who apply there are looking for chances on international contacts.

It can be expected, that the framework conditions for private R&D institutions will not be very good as long as the barrier between academic and industrial staff exists and R&D funding is a stable long-term policy in the country.

Findings: Innovators

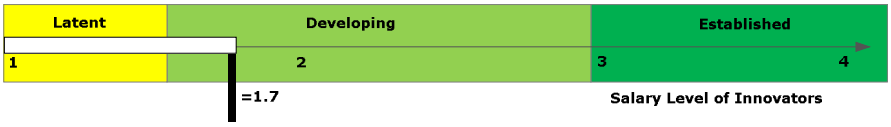


There is a huge potential for innovation in Jordan, many ideas and inventions are generated day by day. Specific support actions, as discussed in previous chapters, would generate a higher number of innovations as well as a more successful commercialisation.

There seems to be a huge number of individuals that are interested in getting more innovation started, although in general the acceptance of these processes in society and politics is low. There are lots of small programmes for support of innovators based on private or public organisations. All these innovators confirmed that friendlier framework conditions would lead to more innovations and better disseminations. They called up the Jordanian government to clearly commitment itself, followed by specific implementing measures and programmes.

Support for innovators is a challenge for all actors within the NIS. It should be based on a clear commitment of the government, followed by specific measures.

Findings: Salary Level of Innovators

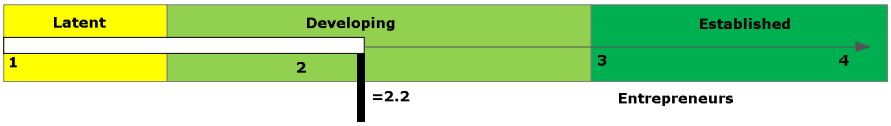


According to the EOS, young people in Jordan are looking for jobs in administration or abroad, nowadays. The rationale seems to be manifold. The salary level of innovators seems not to be that attractive to raise interest.

Salaries of innovators were only a hidden discussion during the interviews. But the experts often mentioned that graduates from universities tend to work in administration bodies or abroad (in neighbouring Arabic countries) due to insufficient salaries in Jordan. Low capacities to do R&D may also be a reason. Therefore, Jordan suffers a high risk of brain drain. In the interviews, our experts often complained about well educated Jordan people that work outside the country.

This problem can only be solved by success stories where innovations lead to personal wealth of the innovator. Increased R&D or innovation capacities may also contribute to attract more young scientist to act locally.

Findings: Entrepreneurs



The social acceptance of entrepreneurs is high and many people are interested in. Thus, this determinant was ranked quite high. The number of entrepreneurs could be increased if the framework conditions and incentives became more attractive.

The number of people that are interested (in theory) in starting their own business is obviously high. There are some examples of success, but this is not enough to convince those, who doubt. The social acceptance for entrepreneurs in Jordan seems to be high as well. The level of management skills is low, but there is a number of training programmes available - for instance by chambers of commerce and Jordan engineers association - but this is much too small with respect to the needs.

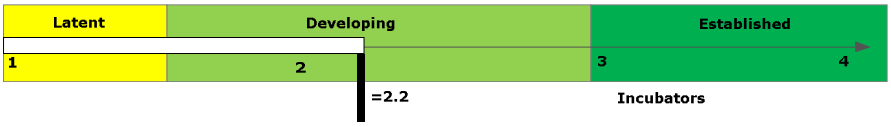
According to the World Bank Country Profile “Doing Business 2010”, Jordan is ranked 100 of 183 in total. But especially the ranking for starting a business (ranking 125) and getting credit (127) is significantly lower. Getting capital and especially risk capital is the major problem for entrepreneurs in Jordan (see chapter private investment). This problem can only be solved by clearly changing the respective policy. If budgets are available, there would be a lot of institutions that could directly and personally support entrepreneurs. Experts from chambers of industry and commerce as well as Jordan Engineers Association declared that these and other organisations are well prepared to do fulfil these tasks.

Although good progress has been made in terms of generating entrepreneurs in Jordan, further public support seems to be recommendable. The needed efforts are comparatively low; the expected impacts are high and reachable in a short time-frame.

Some possible activities were mentioned during the interviews:

- **Entrepreneurial conferences in the regions, in close cooperation with the universities**
- **Awareness measure to motivate more Jordanian citizens to become entrepreneurs**
- **Presenting good examples (local and international) of entrepreneurial business approaches – organised and supported by the patronage of European partners to enhance the credibility**
- **Courses for entrepreneurial skills at all technical universities**
- **Funding programme for entrepreneurs**
- **Accompanying measure to facilitate entrepreneurs and start-ups doing business**
- **Business plan competition**

Findings: Incubators



Incubators are existent and have shown considerable success. Further improvement could be reached if the financial framework was more stable and obtaining capital was facilitated. Additionally, incubators would benefit from stronger political backing.

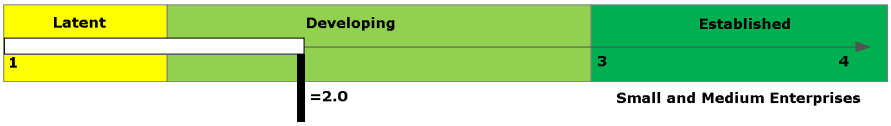
Some (about ten) incubators are existing or under development in Jordan. They are all implemented on activities by different private or public organisations, including EC funding (Support to Research and Technological Development and Innovation Initiatives and Strategies in Jordan - SRTD). During recent years, the existing incubators have shown considerable success. iPark reported having created about 600 jobs since 2003, which can be considered as a good practice. Receiving long-term funding is still a challenge for iPark and the other incubators. As existing incubators are successful, many organisations would prefer to implement new ones, like for instance JEA. But they need at least political commitment and of course stable funding. Nevertheless, Jordanian incubators can be considered as a success story. We are aware that there is a lot of expertise in Jordan how to set up successful incubators and how to train entrepreneurs.

Besides the political and financial background, nearly all experts miss a common "roof" for incubators to gather them in a network and form a kind of lobby to coordinate the work. There are several attempts by institutions that carry the incubators to form a “roof”. But due to the fact, that all of these institutions are in strong competition to reach the limited funding, chances for a tuned approach seem to be low.

Beside ideas of building new incubators, the existing ones are looking for expansion to bring different branches together (networking and synergy) and to keep successful entrepreneurs as mentors in house.

It seems to be advisable to strengthen existing incubators in Jordan rather than to set up new ones. The Jordanian government could implement a clear and stable policy as well as to provide continuous funding on incubators as a central element for the support of entrepreneurship. The return on investment may be excellent. In addition, a strong nation wide network for Jordanian incubators could help to facilitate mutual learning and strengthen their position towards public and policy. Such a network could also increase international visibility to attract international partners and (maybe) investors. This determinant can be a subject of international intervention, since mandatory efforts are low, compared to the expected impact.

Findings: Small and Medium Enterprises (SME)



The Jordanian economy is mainly based on micro enterprises. Only some of them are capable to compete on international level. Thus, Jordanian SME can not be considered as main drivers for innovation.

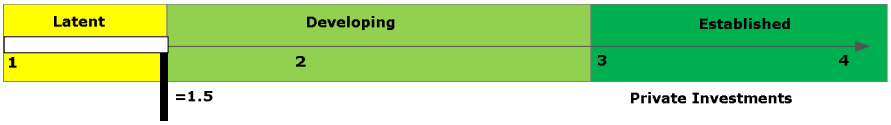
Economy in Jordan is nearly completely based on very small enterprises (micro). According to the Ministry of Industry and Trade, 92 % of enterprises are micro (less then 10 employees[24]), 7.4 % are small and medium-sized (10 to 249 employees) and only 0.6 % are large. The majority of these SME belongs to the trade sector (42,000 enterprises). Most SME are not innovative nowadays. Several international studies see a lack of political support as well as lacks of finance and technological oriented staff. Almost all interviewees agreed, that this is one of the main issues of Jordan's economy and that it is necessary to dramatically increase the number of innovative SME.

As already stated in the chapter "universities", there are only weak links between SME and the academic world. SME do not have the financial resources to have R&D activities on their own. The programme Faculties For Factories is designed for at least medium SME, as financial copayment is required what can hardly be afforded by micro and small SME. The same can be stated for a special programme for graduating students, funded by King Abdullah II. If this programme would focus on SME, it would offer them a chance to gain academic experience for innovation.

[24] according to the definition of the European Commission

A strategic approach to focus on SME capabilities in the country, for example by establishing a powerful SME agency or by delegating this task to existing institutions, could be supported by external expertise and know-how transfer. In the end almost all measures improving the determinants of the meso level would directly improve the innovation capability and competitiveness of the Jordanian SME.

Findings: Private Investments



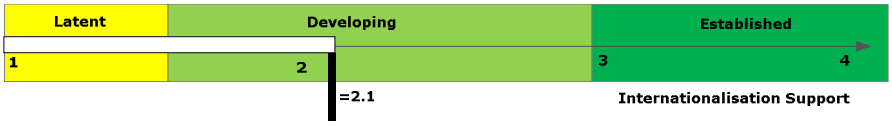
There is a lack of private investments in innovation in Jordan. Especially for entrepreneurs it is very hard to find risk capital. Credits by banks require guarantees that include private savings of complete families, which is normally unacceptable by young people who want to become entrepreneurs or self-employed. That is the reason why this determinant was ranked quite low compared to the others.

It is a main problem for start-up companies and entrepreneurs, that no risk- or venture capital is available. Even "normal" credits by banks require complete guarantees. These guarantees have to be given by the entrepreneur himself or his family. This is a very strong barrier for entrepreneurs.

It is remarkable, that the average score of 1.5 is the lowest one of all scores found in the questionnaires.

The development of the private sector targeting on the support of entrepreneurship and innovation is one of the most challenging tasks for Jordan. The development of the financial sector is outside the focus for possible external intervention, whereas the expected impact on the NIS would be very high.

Findings: Internationalisation Support



According to the interviews, support to redirect Jordan onto the map of international recognition mainly comes from the European Union and Japan. The Medibtikar Euro Med Innovation and Technology Programme is one of the main donors.

Medibtikar[25] is derived from **MED**iterranean and **Ibtikar** (Arabic for Innovation). The Programme's budget is 7.3 million € and it will run till 31st December 2009. The beneficiary countries are: Algeria, Egypt, Israel, Jordan, Lebanon, Morocco, Syria, the Palestinian Authority, Tunisia and Turkey. Some remarkable results are reported as a direct outcome of this initiative. The beneficiary organisations are public and private enterprises concerned with increasing the competitiveness of SME. The Medibtikar project will provide the MEDA countries with new and improved instruments to stimulate innovation at firm and country levels. The project will also simulate networking across the MEDA region and between the region and the countries of the European Union.

Objectives:

- To encourage good practice in **technology and knowledge transfer** by administrations, enterprises, industry federations, chambers of commerce, etc.
- To support the creation and/or improvement of **intermediary organisations** in charge of implementing support policies for SME (Innovation and Technology Centres (ITCs), TechnoParks, incubators)
- To support **innovation management**, from the development of a national innovation strategy to the identification of services to provide to SME
- To provide **support to specific sectors** facing common challenges in the MEDA region

"Support to Research and Technological Development and Innovation Initiatives and Strategies in Jordan" is another EU-funded programme that is located at HCST. The 4 million € programme that started in 2008, is going to boost Jordan's scientific and technological capacities to contribute to the country´s economic development. The programme will be finished by the end of 2009. Extension and redesign is under discussion with the EC.

Jordan should continue the strategy to intensify the relations to international donors and networks. The EU-connection is very strong, especially with the EC, Britain, France and Germany. As a full participant in the Euro-Mediterranean Partnership, Jordan is in a very good strategic position to benefit from international opportunities. Nevertheless, third-donor programmes can not substitute national investments in innovation.

[25] www.medibtikar.eu

5 Scope of Intervention in Jordan

If a NIS should be improved, certain determinants must be addressed. It is quite clear that some determinants are easy to improve, others are much more complex. For this reason, we grouped the determinants according to their complexity for improvement. We used a portfolio with two different scales. One scale represents the "efforts needed" (How extensive is the needed amount of investment to enhance the performance of the determinant?), the other represents the "expected impact" (What range of improvement can be expected?).

To answer the question "What is the scope for intervention?" with highest perspectives for improvement of the capability of the NIS, the results of the interviews and the searched background information are visualised in the following portfolio:

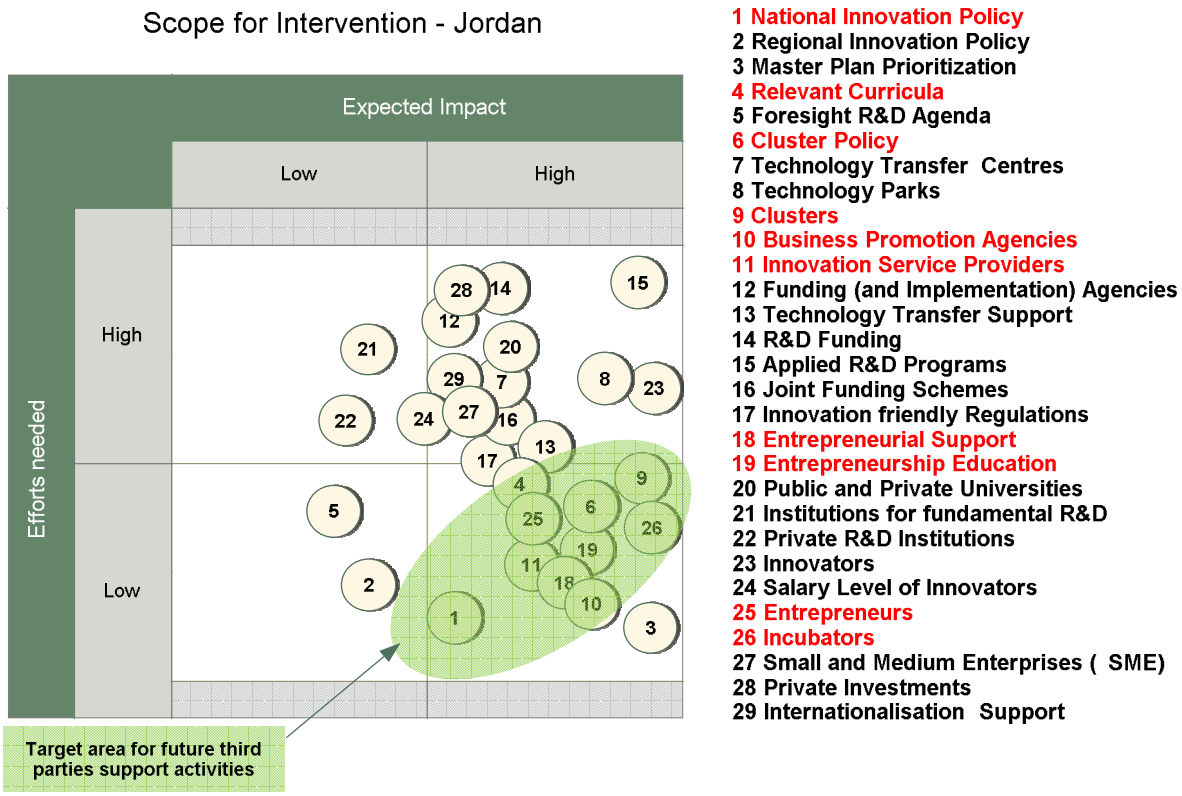


Figure 16 Portfolio – Scope of Intervention in Jordan

The highlighted green area marks the most relevant and possible determinants that could be addressed by foreign third party donors to enhance the NIS. We are convinced that even with limited efforts significant improvements can be gained, provided the recommendations are consequently turned into practise.

The following recommendations describe in first ideas for initialisation and implementation of conceivable activities.

RECOMMENDATIONS FOR ACTION 1

Objective: The national innovation policy provides demand and practically oriented concepts and coordination measures to enhance innovation in Jordan

Jordan needs a nationwide guideline to establish the culture of innovation in all levels of the economy. There is a requirement for modernisation and streamlining of the service framework for the established industry, for new companies, for entrepreneurs and for potential innovators. The developed strategy should cover a policy-measure mix to reach midterm results. Innovative policies must include a clear definition of the importance of modern R&D and state-of-the-art business behaviour for the competitiveness of Jordan. It should be designed as a motivation for young researchers and students to take into account self-employment with the focus on innovative products and services. In result, the national innovation policy should build a proactive strategy involving a wide spread discussion process between all relevant stakeholders, related to academia, research, industry and the support network. The national research strategy developed by HCST can be seen as a nucleus for this process.

A Jordanian national innovation policy can be implemented with little efforts, but high impact. The strategy should be written, stable but adaptable if necessary, independent of persons holding offices or ministries and broadly published. A strong involvement of many important actors from policy, industry and academia is strongly recommended. Funding and legislation must allow innovation processes on all levels to have planning security.

Most important for the success will be the common understanding, the promotion of the policy and its implementation. In the first phase of preparation one ordered ministry of Jordan's government should take leadership in the process. During the future process of implementation and renewing the strategy, more involvement of the intermediaries and support infrastructure will be crucial. This involvement will be achieved easily, as a lot of highly engaged people are in fact waiting for chances to contribute to the national innovation policy. The Jordanian government should decide to support the process by external moderation, regular assessment of distinguished issues and continuous evaluation.

Recommended activities:

- Workshop for national policy makers
  - Good Practice innovation strategies from Europe (Spain, France, Germany)
  - Benchmark with regional approaches
  - Transfer opportunities
- Setting up a high level expert group
  - Jordanian representatives from policy (interministerial), academia and industry
  - Design of a national innovation strategy
  - Some advice/impulse may be given by international experts
- Finding mission and study visit in Germany and other European countries

- Drafting the first innovation strategy
  - Development of the national research strategy already formulated by The Higher Council for Science and Technology
  - Feedback from international expert-network
- Set-up of a communication strategy

Main success factors:

- Strictly orientation towards the national demand
- National policy makers committed to long-term public investments in innovation
- National Policy makers with awareness of the needs of the innovation actors
- National Innovation Strategy focus' on competitive advantages of Jordan
- Clear responsibility for implementation
- Regulatory and fiscal measures fully utilised for innovation

RECOMMENDATIONS FOR ACTION 2

Objective: The innovation support capacities are harmonised and strictly focused on strengthening the modernisation process of industry and research

The management of innovation is one of the main topics for business promotion and innovation service activities. The already existing service provider in Jordan could participate in special designed training measures, enforced by partners and with methods developed through practical experiences. Especially the business promotion agencies and the innovation service providers could benefit from these activities. Discussions with stakeholders from these institutions made clear that attendance for higher engagement can be expected.

To achieve sustainable results, all stakeholders in Jordan with relation to technology, business development and innovation have to bundle their individual strengths to an efficient and powerful workforce. There are already excellent ongoing examples of support activities in Jordan. The Jordan Engineers Association for example already has a wide range of innovation support activities, which could easily be extended.

Even the policy-framework is more and more related towards the modernisation process of the economy. Still missing is the coordinated approach to avoid inefficiency and misunderstandings. To provide the basis for business development, better synergy between industry and academia is needed.

Technology transfer support – the implementation of a stable bridge between industry and universities or research institutes is needed to enable absorptive capacities of technology and innovation. Good practice examples from Europe and neighbouring countries could help to build up a strategy with mid-term impact. Activities like Europe INNOVA are suitable benchmarks to learn from.

The Jordanian government should give stable political background and funding to innovation service providers. Significant achievements with comparable low efforts are expected as the infrastructure and the actors already exist. With small budgets, for management and entrepreneurship training for example, high impact can be expected. Setting up clear programmatic framework conditions would help as well.

Recommended activities:

- Coordination by high level expert group
  - Setting-up of sub-groups
- Survey of existing nationwide support capacities – transparency
  - Avoiding duplications
- Identification of funding gaps
  - Finding mission and study visit in Germany and other European countries
- Adjustment of existing measures

Main success factors:

- Involvement of the main actors in the Innovation System
- Willingness for transparency and cooperation
- Coordinated communication
- Break-up of traditional systems and responsibilities, if necessary

RECOMMENDATIONS FOR ACTION 3

Objective: The culture of entrepreneurship is acknowledged as one of the main elements to achieve economic development in Jordan

Entrepreneurship is about creative people, their options and activities in starting, taking over or running a business. Entrepreneurship can occur in any sector and type of business. It applies to the self-employed and to firms of any size throughout the various stages of the business-life-cycle, from pre-start to growth, transfer and re-start. Entrepreneurship is relevant for firms in all sectors and technology-fields.

The promotion of entrepreneurial skills and willingness, to enhance individual's motivation and enthusiasm is a mission for public authorities. The long term impact for the society should build the driving factor for political activities. In Jordan, all public activities should assemble a combination of the main intellectual and economical forces of the society. Entrepreneurship needs the influence and the interaction of research (universities), education (schools), networking (associations, e.g. of engineers) and business opportunities (private sector, national and international customer). The role of the state is visible in the responsibility for growth- and stability oriented strategies and macro-economics policies. The aims of such public activities are

- Motivating and educating interested people to build up own enterprises
- Allocating the needed infrastructure, like commercial premises
- Enhancing the acceptance of innovation and the strove for independence

Entrepreneurship support should come into focus of the Jordanian policy, since there are good results gained so far. In the transition phase towards a social market economy in Jordan there is a growing need for new enterprises and for the common behaviour of risk-taking among Innovators. To build up the culture of entrepreneurship the existent support activities could be enhanced by international knowledge-transfer and funding programmes offered by the Jordanian government. Nevertheless, since entrepreneurs are considered to act as key drivers for innovation, the number of entrepreneurs in Jordan should be increased. The design and the implementation of these programmes could be supported by practically experienced partners from Europe. There are very successful support measures for all levels of entrepreneurial awareness and development in different European countries which could be used as practical benchmarks. Especially in the land-scape of academia in Jordan, entrepreneurial education could be extended. The skills among the teaching staff seem to be expandable by directly addressed activities, like train-the-trainer coaching. A solution for allocation of more risk capital and a credit system with acceptable guarantees is a precondition for success.

Further support measures are considered to be essential to further improve the entrepreneurial surrounding. Stable funding conditions should be provided to existing and new actors in order to enable mid-term projects. Implementing smart programme design features would help as well. The Jordanian government could implement a clear and stable policy as well as to provide continuous funding on incubators as a central element for the support of entrepreneurship. The return on investment may be excellent.

Recommended activities:

- Conference “Entrepreneurship and Innovation Culture”
  - Lectures and practical workshops
  - Participants from Jordan, neighbouring countries and Europe
- Coaching programme for innovation support agencies – “how to enhance the entrepreneurial culture and how to raise awareness among the young generation”
  - Participants: business promotion agencies, technology transfer and entrepreneurship supporter
- Training session “Entrepreneurial Education”
  - Practical excursions to European partners (like career services) and to establish (twinning approach) close relations with them
  - Participants: universities, technology transfer and entrepreneurship supporter
- Awareness campaigns in universities
- Adjustment of existing measures

Main success factors:

- Involvement of the main actors in the field of entrepreneurship support
- Willingness for transparency and cooperation
- Coordinated communication
- Openness for international cooperation and good-practice-learning

RECOMMENDATIONS FOR ACTION 4

Objective: Incubators and incubation services are enhanced in Jordan to enable the transition of innovative ideas into competitive products and services

Business incubators as the “infrastructural element” of a NIS are a success story for many years all over the world and recently in several MENA countries. The approach of a secure environment for entrepreneurs, enriched with all needed services and networks still seems to be an enabler for business development and economic fortune. The idea has evolved over the time and has special characteristics depending on the local conditions. Strategies of branch-oriented incubators or women incubators are more common these days. The challenge for governments and development agencies is to develop capable and cost-effective solutions to create and finance incubator initiatives.

In many countries the process of structural reforms is still ongoing or at the beginning. To learn from good practice examples in the field of business incubation could be a helpful approach to avoid failures and time-lacks.

In Jordan several incubators have been established and shown good results. There is still a significant willingness for many institutions to create even more incubators across the country. But Jordan needs to establish a promising incubation framework providing a stable financial basis, a common quality management and permanent evaluation. Nevertheless, it seems to be advisable to improve the performance of existing incubators rather than to create more. Dedicated focus should be given to improve the services and added-value provided by the incubators as well as to improve the framework conditions (increased public invests, entrepreneurial-friendly framework condition, decreased administrative hurdles, etc.). In addition, a strong nation wide network for Jordanian incubators could help to facilitate mutual learning and strengthen their position towards public and policy. An actually visible network could also increase international visibility to attract international partners and (maybe) investors.

Recommended activities:

- Deeper analysis of the current situation in Jordan
  - SWOT-Analysis, interviews and site visits
  - Recommendations for the strategy process
  - Participants: incubation management, innovation promotion agencies, technology transfer and entrepreneurship supporter
- Training concerning the design of business models for incubators
  - Recommendations for potential approaches to establish a sustainable business framework, mentioning environment strengths and weaknesses by an experienced European partner out of the practical incubation business

- Leadership-training for existing and future managers of incubation facilities
  - Supporting the involvement of Jordan in the MENAinc Network (World Bank)
  - Performance by: experienced Incubator Manager from European and the MENAinc Network
  - Twinning-projects

RECOMMENDATIONS FOR ACTION 5

Objective: The innovation policy in Jordan is focused towards cluster creation in future-driven sectors (like renewable energy, ICT and pharmacy)

The clustering of partners and activities to focus on selected branches with high potential for the regional and international competitiveness is one of the most prospective strategies for Jordan. Clusters and networks are regional concentrations of interconnected universities, research institutions, service providers, small and large enterprises, educational institutions, public actors and further associated institutions in a particular field. They comprise the various steps of the value-added chain and are characterised by an intense interaction and communication among the involved players. Enterprises benefit from the research results at an early stage and likewise research institutions find business partners for implementing their research results more effectively.

Thus, clusters and networks are important as well as effective instruments for the promotion of innovation, development of economy and business as well as location marketing. Clusters around pharmaceutical industry and ICT exist in Jordan, but could generate increased leverage when sufficiently supported by public authorities.

Key success factor for the cluster performance and the increased innovation capability and competitiveness of the cluster firms are demand oriented and tailor-made services offered by a cluster coordinating body. This may be the focus of future interventions and support. Analysing the potentials for future would-be clusters as well as the demand for services and added-values provided by clusters would be the first step. It seems to be promising to set up cluster management entities to continuously support the cluster members in specific fields. These cluster management entities and their staff must receive a kick-off financing from the Jordanian state.

Recommended activities:

- Deeper analysis of the capabilities, especially on the assessment of the institutional contexts
  - Identification of would-be clusters (nucleus of clusters) or regional networks
- Design of strategy, technological focus and main services instruments
  - Recommendations for potential approaches to establish a sustainable cluster approach
  - Practical guide “how to build a cluster initiative”
- Leadership-training to build personal capacities for cluster management
  - Train-the-trainer approach
  - Performance by: experienced cluster manager from Europe
  - Twinning-projects
- Implementation of cluster approaches
  - Coaching by external cluster experts

6 NIS of Syria, Jordan and Egypt - Comparative Overview

In the former chapters we conducted a comprehensive analysis of the 29 determinants as a status-quo assessment of the Jordanian NIS. It was the main intention to characterise the determinants and to identify those that may be positively affected by third parties in the future.

This chapter is intended to present a comparative overview of the Syrian, Jordanian and Egyptian determinants. It is not our intension to rank these three NIS rather than to reveal country-specific strengths of certain determinants which may initiate a mutual learning among each other. Such benchmarking becomes more and more relevant in many fields. We selected Egypt for the benchmarking since its NIS is reported to be quite advanced compared to the most nations of the North African and Middle Eastern regions. Further more, we run an exploratory study on the Egyptian NIS in 2008 which gave us some value insight views[26]. Since the assessment of the Egyptian NIS was not in the focus of the Term of Reference, we were not able to run interviews with a comparable number of experts like in Jordan. Thus, the values of the determinants do not have the same validity like for Jordan, but we are convinced that they give a reliable picture.

Macro Level – Policy Level

The findings related to the determinants belonging to the macro level/policy level reveal that there seems to be no considerable difference between those three countries. The patterns are well balanced; the respective values are all around 1.5 and 2.0. This means the innovation related policies of all three countries seem to have comparable characteristics, or in other words, have the same strengths and weaknesses. In all countries there is no national, harmonised innovation policy available. Often some respective elements are regarded in certain plans and official documents. In addition, regional innovation policies or master plans for turning such policies into practice are not common. For all three countries, cluster policies do not really exist, although in Jordan some bottom-up clusters have emerged in the recent past. Finally, it seems to be worth mentioning, that the determinants for Jordan continuously ranked higher compared to the other countries.

[26] Meier zu Köcker, G; Hahn, P, The National Innovation System in Egypt – An exploratory study of Egyptian Technology Transfer Centres, financed by GTZ, www.iit-berlin.de/veroeffentlichungen, 2008

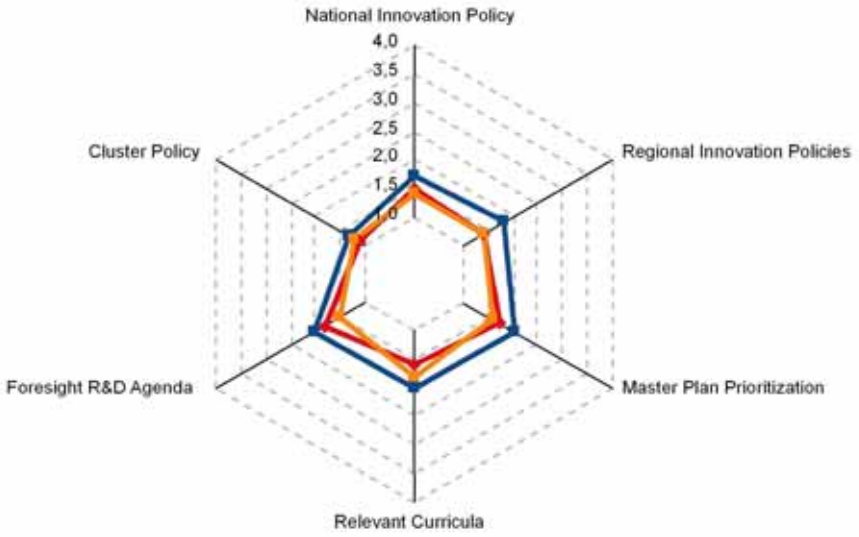


Figure 17 Comparison of the determinants related to the policy level (macro) between Syria (red), Jordan (blue) and Egypt (orange)

Meso Level – Institutional Innovation Service Providers

In the previous chapters we stated that institutional service providers are main tools to turn innovation policy into practice and often the main source to support the actors of a NIS in innovation matters. Thus, the characteristics of the respective determinants are quite important and reveal how innovation policy or official commitments are implemented in reality.

The determinants for Egypt are continuously ranked higher as for Syria and Jordan, except for clusters. The latter one does not really exist in Egypt, whereas in Jordan some clusters do exist. On the other hand, technology transfer centres are much more developed and receive stronger public support in Egypt compared to the other countries. The same can be seen for funding agencies and innovation service providers. The latter ones do receive better public support and provide demand-oriented services. Nevertheless, there is room for improvements even for the Egyptian institutional innovation service providers. As far as business promotion agencies are concerned, determinants of all countries reach comparable values. Although there a more technology parks existing in Egypt than in the other countries, this determinant was not ranked very high by the Egyptian interviewees since they often lack to provide real added values. Smart Village Egypt is reported to be a very positive exception. In some cases there are only buildings and no services are provided for the tenants.

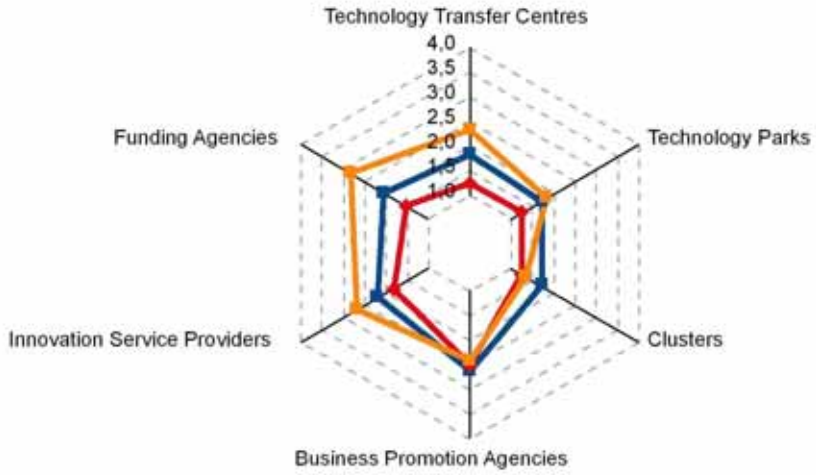


Figure 18 Comparison of the determinants related to the institutional innovation support level (meso) between Syria (red), Jordan (blue) and Egypt (orange)

Meso Level – Programmatic Innovation Service Providers

Programmatic innovation service providers are the second backbone of a national innovation support system. The patterns of the three countries are quite comparable. All three are well balanced with maxima on entrepreneurial support. The determinants for public funded programmes (R&D funding, applied R&D funding and joint funding schemes) are ranked lower, compared to the others. Here similar low values are reached, since the governments hesitate to significantly invest in public programmes supporting science, research and innovation. Nevertheless, Egypt invests much more in technology transfer schemes, not related to R&D. Entrepreneurial support is of high relevance in all countries, whereas the determinants of Egypt are ranked very high, compared to the others. This seems to be a clear outcome of significant efforts the Egyptian government has been spending for several years. Finally, we very surprised about the finding that the Syrian determinants are often close to those of Jordan.

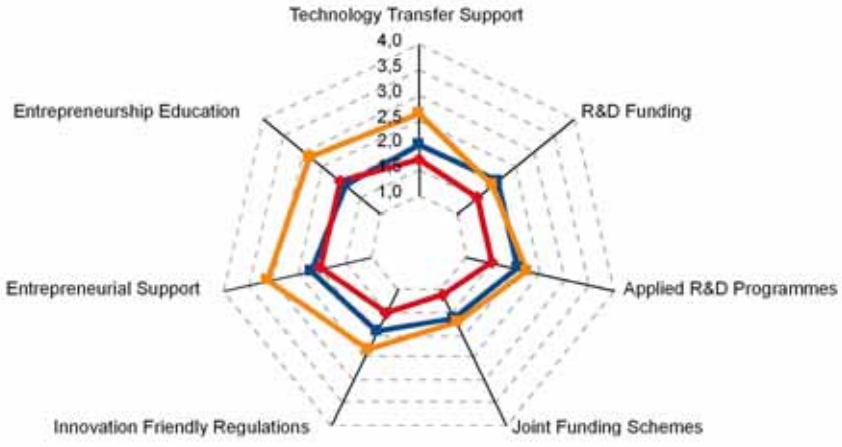


Figure 19 Comparison of the determinants related to the programmatic innovation support level (meso) between Syria (red), Jordan (blue) and Egypt (orange)

Micro Level – Innovation Capacity

The innovation actors of a NIS are the main drivers for innovation, strongly depending on the framework conditions provided by the respective governments. Figure 20 reveals the respective patterns. All three countries have different actors who have been ranked higher than in other countries. As far as Syria is concerned, innovators, entrepreneurs as well SME are assessed to be on the same level as in Egypt and Jordan. Universities and international support actions are considered to be most developed in Jordan. Private R&D institutions and investors seem to have the best conditions in Egypt, whereas the absolute values are not that high.

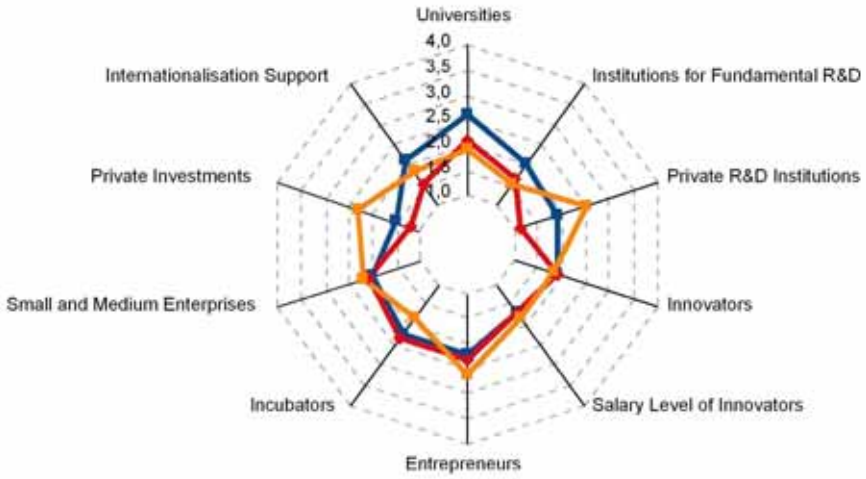


Figure 20 Comparison of the determinants related to the innovation capacity level (micro) between Syria (red), Jordan (blue) and Egypt (orange)

According to our perspective the findings clearly point out where specific countries have individual strengths and weaknesses. If policy makers or interested parties are keen to learn from good practice, there is no need to look always at Europe rather then to learn from the neighbours. For sure, in some areas, like public programme-based support initiatives, all three countries should gain experiences from other countries.

7 Appendix I: Programme-Descriptions and Other Related Documents

www.jei.org.jo

Jordan Education Initiative

Education is recognized as a key catalyst and enabler for social and economic development in countries throughout the world. Recognizing this, business leaders from the IT and telecommunications sectors at the World Economic Forum Annual Meeting 2003 presented the World Economic Forum with a mandate to launch an action-oriented initiative to leapfrog education reform in a selected pilot developing country using a new framework for public-private collaboration and partnership.

Jordan, with a committed, clear, and aggressive strategy for education reform was selected as the pilot country for the learning initiative. Formally launched in June 2003 at the Extraordinary Meeting of the World Economic Forum at the Dead Sea, the Initiative aims at accelerating education reform through a public-private partnership model that drives innovation and capability.

The Initiative was developed in partnership between the Government of Jordan and the World Economic Forum and to date incorporates over 17 global corporations, 17 Jordanian entities, and 11 governmental and non-governmental organizations as stakeholders in achieving its goals.

The Initiatives objectives are to:

- Improve the development and delivery of education to Jordans citizens through public-private partnerships, and in the process help the government of Jordan achieve its vision for education as a catalyst for social and economic development
- Encourage the development of an efficient public-private model for the acceleration of educational reforms in developing countries based on unleashing the innovation of teachers and students through the effective use of ICT
- Build the capacity of the local information technology industry for the development of innovative learning solutions in partnership with world class firms, creating economic value that will lead to mutually beneficial business opportunities
- Leverage an environment of national government commitment and corporate citizenship to build a model of reform that can be exported to and replicated in other countries

Jordan Innovation Centre Network Quality System

New approach, inspired by the European EBN Quality System

Entrepreneurship Promotion at Educational Institutions: A Model Suitable for Emerging Economies

Dr. Bashar S. El-Khasawneh, Assistant Professor

Industrial Engineering Department

College of Engineering

Jordan University of Science and Technology

Abstract: - This paper discusses the role of educational institutions in promoting entrepreneurship and risk taking, particularly in conservative communities and cultures. Two main topics are discussed in this paper. The first is tailoring the academic program and all activities supporting it to serve the goal of promoting entrepreneurship. The second topic is to construct a particular form of incubators suitable for emerging economies which is financial resources bound. These incubators serve multiple functions, therefore, in addition to the traditional role of incubating potential-businesses; the other roles are serving as a training site for business-related and soft skills topics for students and serving as an expert house for the industry. This would promote the universities' role in serving and prospering their communities impacting positively the emerging economy. Efforts related to these topics by a Jordanian University will be discussed as the case study.

8 Appendix II: Literature

General Information

1

European Commission (Directorate-General for Enterprise and Industry); Organisation for Economic Cooperation and Development (Private Sector Development Division); European Training Foundation: Report on the implementation of the Euro-Mediterranean Charter for Enterprise, 2008 enterprise policy assessment, in consultation with: European Investment Bank; © European Communities/OECD, 2008

2

World Economic Forum within the framework of the Global Competitiveness Network: The Global Competitiveness Report 2009-2010, © 2009 World Economic Forum

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Niels Bosma, Zoltan J. Acs, Erkko Autio, Alicia Coduras, Jonathan Levie: Global Entrepreneurship Monitor, 2008 - Executive Report, 2009

Jordan

4

Jordan Economy 2009; http://www.theodora.com/wfbcurrent/jordan/jordan_economy.html, source: 2009 CIA World Fact Book

5

*info*Dev: International good practice for establishment of sustainable IT parks; An publication prepared by: PriceWaterHouseCoopers, India, June 2008; ©2008 The International Bank for Reconstruction and Development/The World Bank

6

Jordan Tourism Cluster; prepared for: Microeconomics of Competitiveness; May 8, 2009

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European Training Foundation: Technical and Vocational Education and Training in Jordan, Areas for development cooperation; External Communication Unit, *2006*

8

USAID: R&D Assessment at Jordanian Universities: Encouraging Transfer of Technology to Innovation Stage; Jordan Economic Development Programme (SABEQ)

9

EUROPEAN NEIGHBOURHOOD AND PARTNERSHIP INSTRUMENT, Jordan Strategy Paper 2007 - 2013 & National Indicative Programme 2007 – 2010

10

David Tee: A Practical Guide to Wealth Creation, Proposals for an Innovation Policy for Jordan, Euro-Jordanian action for the development of enterprise (EJADA), 2006

9 Appendix III: List of Interview Partners

Jordan

Institution
Ministry of Higher Education and Research
Ministry of Planning & International Cooperation
Ministry of Industry & Trade
Ministry of Energy and Mineral Resources
Jordan's Technology Incubator
SRTD (Support to Research & Technological Development & Innovation Initiatives and Strategies in Jordan)
The Higher Council for Science and Technology
Amman Chamber of Commerce
German-Jordanian University
Jordan Enterprise Development Corporation
Development & Employment Fund
Jordan Engineers Association
Jordan Forum for Business & Professional Women
El Hassan Science City
Royal Scientific Society
Queen Rania Centre for Entrepreneurship
Amman Chamber of Industry
King Abdullah II Design and Development Bureau (KADDB)
Jordan Innovation Centre for Engineers & Industrial Enterprises at Royal Scientific Society
Consultant, Hiba Al-Ali
I's Of Innovation
Envision Consulting Group
King Abdullah II Fund for Development
SABEQ (USAID Jordan Economic Development Program)

Institution: Ministry of Higher Education and Research
Name: Yasser Mismar (and Ms Eman Al-Atom)
Position: Scientific Research Dep. Director
Date of Interview: 2nd of August 2009

Institution: Ministry of Planning & International Cooperation
Name: Dr. Saleh Kharabsheh (and Philipp Godron, CIM Expert)
Position: Project Dep. Director
Date of Interview: 3rd of August 2009

Institution: Ministry of Industry & Trade
Name: Eng. Moh'd Khaldi
Position: Head of SME Division
Date of Interview: 3rd of August 2009

Institution: Ministry of Energy and Mineral Resources
Name: Zeyad Gbreel
Position: Renewable Energy Dep. Director
Date of Interview: 3rd of August 2009

Institution: Jordan's Technology Incubator
Name: Dr. Wesam Rabadi
Position: GM
Date of Interview: 3rd of August 2009

Institution: SRTD (Support to Research & Technological Development & Innovation Initiatives and Strategies in Jordan)
Name: Dr. Raghda Kilani
Position: Director
Date of Interview: 3rd of August 2009

Institution: The Higher Council for Science and Technology
Name: Falak H. Sarraf
Position: Director of Industrial Scientific R&D Fund
Date of Interview: 3rd of August 2009

Institution: Amman Chamber of Commerce
Name: Mohannad Al Attar
Position: General Manager / Committee member
Date of Interview: 4th of August 2009

Institution: German-Jordanian University

Name: Dr. Hisham Gharaibeh

Position: Dean of Talal Abu Ghazaleh School of Business

Date of Interview: 4th of August 2009

Institution: Jordan Enterprise Development Corporation

Name: Hana Uraidi

Position: Director Cross Cutting Support Directory

Date of Interview: 4th of August 2009

Institution: Development & Employment Fund

Name: Ali Ghezawi

Position: Director General

Date of Interview: 5th of August 2009

Institution: Jordan Engineers Association

Name: Abdallah Obaidat (and Khaled Qadoomi, Assistant Secretary General and Nasser M. Huneidi, Secretary General)

Position: President of JEA

Date of Interview: 5th of August 2009

Institution: Jordan Forum for Business & Professional Women

Name: Rania Khatib

Position: Chief Executive Officer

Date of Interview: 5th of August 2009

Institution: El Hassan Science City

Name: Omar Hamarneh

Position: Chief Executive Officer

Date of Interview: 5th of August 2009

Institution: Royal Scientific Society

Name: Khaled Kahhaleh, Ph.D

Position: Vice President

Date of Interview: 5th of August 2009

Institution: Queen Rania Centre for Entrepreneurship

Name: Mohamad M. Khawaja (and Mohammed Kilani, Programme Coordinator)

Position: Executive Director

Date of Interview: 5th of August 2009

Institution: Amman Chamber of Industry

Name: Dr. Ibrahim Ali Rewabdeh (and Fadel Labadi, Manager of Indsutrial Affairs Depart.)

Position: Chief Executive Officer

Date of Interview: 6th of August 2009

Institution: King Abdullah II Design and Development Bureau (KADDB)

Name: Dr. Bashar S. El-Khasawneh

Position: Director of Applied Research Programmes

Date of Interview: 6th of August 2009

Institution: Jordan Innovation Centre for Engineers & Industrial Enterprises at Royal Scientific Society

Name: Nizar Al-Halasah

Position: Director

Date of Interview: 6th of August 2009

Institution: Consultant

Name: Hiba Al-Ali

Postition: PhD Candidate at McGill University, Canada

Date of Interview: 4th of August 2009

Institution: I's Of Innovation

Name: Yousef Abu-Hmeidan

Position: Consultant

Date of Interview: 2nd of August 2009

Institution: Envision Consulting Group

Name: Yusuf Mansur, PhD

Position: CEO

Date of Interview: 3rd of August 2009

Institution: King Abdullah II Fund for Development

Name: Imad Yared

Date of Interview (filled questionnaire)

Institution: SABEQ (USAID Jordan Economic Development Program)

Name: Laith Al Qasem

Position: CEO

Date of Interview (filled questionnaire)